



Saikosaponin C ameliorates tau-related pathology by modulating oxidative stress and MAPK axis in Alzheimer's disease

Dongmei Chen^{a,1}, Ruomeng Li^{a,1}, Fei She^a, Jiawen Chen^a, Sang Min Cho^b, Bohyun Yang^b, Yuan Tian^a, Le Guan^c, Xiuzhi Zheng^a, Quling Wang^a, Long Wang^a, Sang Won Park^{d,e}, Wucheng Tao^{a,c}, Tao Zhang^{a,*}, Byeong Mo Kim^{b,**}, Tae Ho Lee^{a,***}

^a Fujian Key Laboratory of Translational Research in Cancer and Neurodegenerative Diseases, Institute of Basic Medicine, School of Basic Medical Sciences, Fujian Medical University, Fuzhou, Fujian, 350122, China

^b Research Center for New Drug Development, AgingTarget Inc., Gwachen-si, Gyeonggi-do, 13840, Republic of Korea

^c Key Laboratory of Brain Aging and Neurodegenerative Diseases, School of Basic Medical Sciences, Fujian Medical University, Fuzhou, Fujian, 350122, China

^d Division of Endocrinology, Boston Children's Hospital, Boston, MA, 02115, USA

^e Department of Pediatrics, Harvard Medical School, Boston, MA, 02115, USA

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ABSTRACT

Ethnopharmacological relevance: Saikosaponin C (SSc), a traditional Chinese herbal medicine, is a triterpene saponin and an active compound extracted from *Radix Bupleuri*. It has been demonstrated to have neuro-protective effects in cellular models of neurodegenerative diseases; however, its precise mechanism of action in alleviating tau pathology in Alzheimer's disease (AD), as well as its potential to improve cognitive deficits in animal models, has not yet been elucidated.

Aim of the study: In this study, we investigated the *in vivo* therapeutic effects of SSc on a human tau (hTau) mouse model expressing normal hTau isoforms in terms of tau-related pathology (tauopathy) and the underlying mechanisms.

Materials and methods: For the animal study, C57BL/6 mice received stereotaxic brain injections of adeno-associated virus (AAV)-hTau in the absence or presence of SSc. The Morris water maze test was employed to evaluate the therapeutic effects of SSc on memory and learning. Tau phosphorylation in the brain was assessed by immunofluorescence and Western blotting. Dendritic spine maturation, an indicator of synaptic plasticity, was determined by assessing doublecortin expression and performing automatic dendritic spine analysis. GFAP and Iba1 immunoreactivity were observed as indicators of neuroinflammation. Dityrosine formation and superoxide dismutase 2 expression were measured to monitor oxidative stress. For the mechanistic study, RNA-Seq was used to identify tauopathy-related changes in gene expression. Furthermore, the effects of SSc on aluminum chloride (AlCl₃)-induced cytotoxicity, oxidative stress, tau phosphorylation, and mitogen-activated protein kinase (MAPK) signaling were evaluated.

Results: We found that SSc significantly relieved spatial memory impairment and alleviated the pathological hallmarks of AD-like tau, such as excessive tau phosphorylation and oxidative stress, in AAV-hTau-injected mice. SSc also relieved the deficit in dendritic spine density in mice with tauopathy-associated dementia. Neuro-inflammation, another hallmark of AD-like pathology, was reduced by SSc treatment. Furthermore, SSc attenuated AlCl₃-induced tau pathology, such as neurotoxicity and abnormal tau phosphorylation, by targeting oxidative stress and the downstream MAPK pathway.

* Corresponding author. Fujian Medical University, 1 Xuefu North Road, Fuzhou, Fujian, 350122, China.

** Corresponding author. Research Center for New Drug Development, AgingTarget Inc., A202, URBAN HUB, 60, Gwachen-daero 7na-gil, Gwachen-si, Gyeonggi-do, 13840, Republic of Korea.

*** Corresponding author. Fujian Medical University, 1 Xuefu North Road, Fuzhou, Fujian, 350122, China.

E-mail addresses: dmchen88@fjmu.edu.cn (D. Chen), hupimao@live.com (R. Li), Fei.She@outlook.com (F. She), 18150021261@163.com (J. Chen), smcho@agingtarget.com (S.M. Cho), byang@agingtarget.com (B. Yang), tian18615526627@163.com (Y. Tian), leguan7647@163.com (L. Guan), zhangtailu66666@163.com (X. Zheng), quling502959@163.com (Q. Wang), wanglong@fjmu.edu.cn (L. Wang), SangWon.Park@childrens.harvard.edu (S.W. Park), taowucheng@fjmu.edu.cn (W. Tao), taozh@fjmu.edu.cn (T. Zhang), bmkim@agingtarget.com (B.M. Kim), lee0813@fjmu.edu.cn (T.H. Lee).

¹ These authors contributed equally to this work.

Conclusions: Our results suggest that SSc treatment ameliorates cognitive deficits and related tau pathological features in the hTau AD mouse model. The antioxidant effects of SSc might be responsible for the therapeutic potential of SSc in protecting against tau pathology and cognitive decline.

1. Introduction

Tau-related pathology (tauopathy), characterized by intraneuronal neurofibrillary tangles of paired helical filaments, is one of the major age-related neurodegenerative disorders, including Alzheimer's disease (AD) (Rawat et al., 2022; Wang and Mandelkow, 2016). Elevated phosphorylation and subsequent aggregation of tau are considered the main causes and hallmarks of tauopathy (Rawat et al., 2022). To date, approximately 50 phosphorylation sites have been identified in hyperphosphorylated tau in humans with AD (Wegmann et al., 2021). The phosphorylation of the tau protein is catalyzed by protein kinases, whereas phosphatases reduce tau phosphorylation levels (Ferrer et al., 2005; Morris et al., 2011). Dysregulation and dysfunction of either protein kinases or phosphatases contribute to tau pathologies in AD, and restoring the equilibrium between protein kinases and phosphatases is a promising avenue for AD therapy (Singh et al., 2024). Our previous studies revealed that inhibition of protein kinase activity significantly reduces the phosphorylation level of tau protein and alleviates tau-related lesions (Chen et al., 2020, 2022; Zhang et al., 2024a, 2024b). The progression of tau pathology is closely related to the degeneration of brain function and structure, as well as the decline in cognitive function (Bejanin et al., 2017; Neddens et al., 2018; Rui et al., 2025; Zhou et al., 2023). Transgenic mice overexpressing human tau (hTau) in the brain are widely used as models for human tauopathies, such as AD, to study tau-associated processes, including neurotoxicity, neuroinflammation, synaptic loss, neurodegeneration, and cognitive decline (Andorfer et al., 2003; Duff et al., 2000; Yin et al., 2016). These mouse models exhibit the systemic spread and propagation of tau pathology in the brain, with histological and ultrastructural characteristics similar to those of authentic tau lesions. Recent studies have shown that several drugs targeting tau, such as the monoclonal antibody bepranemab, the monoclonal antibody JNJ-63733657 and the small molecule inhibitor LY3372689, are in phase 2 clinical trials (Huang et al., 2023). However, the long-term safety and efficacy of these drugs remain to be further studied.

Oxidative stress has been reported to be a causal event in AD (Bai et al., 2022; Mecocci et al., 2018). Many studies have demonstrated that the levels of markers of oxidative damage to intracellular macromolecules, as well as reactive oxygen species (ROS), are significantly increased in the brain and plasma of patients with AD or mild cognitive impairment (MCI) (Ionescu-Tucker and Cotman, 2021; Mecocci et al., 2018). A growing body of research has shown that tau hyperphosphorylation promotes the occurrence and development of oxidative stress, and oxidative stress also plays an important role in tau pathologies in AD (Campos et al., 2022; Denechaud et al., 2023; Promyo et al., 2020; Xu et al., 2020). Several studies have shown that the overexpression of tau protein can increase the level of oxidative stress (Denechaud et al., 2023; Xu et al., 2020). Furthermore, oxidative stress induced by aluminum chloride (AlCl₃) was reported to increase the levels of tau phosphorylation both *in vivo* and *in vitro* (Campos et al., 2022; Promyo et al., 2020). The inhibition of oxidative stress can alleviate tauopathies and cognitive dysfunction (Fisar and Hroudova, 2024; Xu et al., 2020).

Recent studies increasingly indicate that traditional Chinese medicine may alleviate symptoms associated with AD (Yan et al., 2025; Zhou et al., 2024). Derived from the dried roots of *Bupleurum chinense* DC or *Bupleurum scorzonrifolium* Willd., *Radix Bupleuri* has been utilized in traditional Chinese medicine for more than 2000 years (Qu et al., 2022). A substantial body of research has demonstrated the therapeutic potential of *Radix Bupleuri*, showing its antioxidant, anti-inflammatory,

anticancer, antiviral, hepatoprotective, neuroprotective, and immunomodulatory properties (Jia et al., 2019; Jiang et al., 2025; Lei et al., 2025; Li et al., 2022; Yang et al., 2017, 2024). It is predominantly cultivated in the subtropical regions of the Northern Hemisphere (Qu et al., 2022) and has traditionally been used to treat conditions such as cold, fever, inflammation, depression, and liver diseases (Feng et al., 2020; Ni et al., 2009; Wang et al., 2023). The earliest recorded use of this plant appeared in Shen Nong Ben Cao Jing, a foundational text in traditional Chinese medicine (Jia et al., 2022; Zhong et al., 2024). The classical herbal formulation "Xiaoyaosan", documented in traditional Chinese medical literature, incorporates *Radix Bupleuri* as a principal component synergistically combined with complementary botanicals, demonstrating clinically significant antidepressant efficacy in pharmacological studies (Yang et al., 2017). *Radix Bupleuri* is known to contain a variety of bioactive compounds, including saikosaponins, polysaccharides, flavonoids, sterols, and organic acids (Zhong et al., 2024). Among the saikosaponins, several subtypes have been identified, such as saikosaponin A, B1, B2, B3, B4, C, D, E, F, G, H, and I (Han et al., 2024). Saikosaponin C (SSc) contains three sugar chains attached to the aglycone (Supplementary Fig. S1). While this compound is known to promote endothelial growth and angiogenesis, it does not exhibit anticancer effects (Okita et al., 1993; Shyu et al., 2004). The therapeutic effects of SSc on liver disease and sepsis have also been reported (Lee et al., 2014; Pan et al., 2019). Recent studies have shown that SSc, recognized as an active ingredient, can alleviate depression-like behavior by suppressing inflammation (Bai et al., 2023). Our previous study revealed that SSc exerts neuroprotective effects by inhibiting abnormal tau phosphorylation, thereby enhancing dendritic spine and synapse formation in neuronal cell models (Lee et al., 2016). However, the underlying mechanisms by which SSc mitigates tau pathology, as well as its potential to improve cognitive deficits in animal models, have yet to be investigated.

In the present study, we aimed to investigate the therapeutic effects of SSc on tauopathy *in vivo*. We induced adeno-associated virus (AAV)-mediated tau pathology in wild-type (WT) mice. Our findings indicate that SSc provides neuroprotection against AAV-tau toxicity in mice by inhibiting pathological tau phosphorylation and reducing neuroinflammation. We also found that the antioxidant properties and ability of SSc to regulate MAPK signaling may provide therapeutic advantages for the potential treatment of tauopathy.

2. Materials and methods

2.1. Drugs and reagents

SSc (purity 98%) was purchased from Herbest (Baoji, Shanxi, China). For the animal experiments, SSc was used to generate an SSc diet (0.0125%). For the cell experiments, SSc was dissolved in DMSO to a stock concentration of 10 mM. DMSO (0.2%) was used as a vehicle control. AlCl₃ and antioxidants, such as N-acetylcysteine (NAC) and 4-hydroxy-2,2,6,6-tetramethylpiperidine-1-oxyl (Tempol), were purchased from MilliporeSigma (St. Louis, MO, USA). MAPK inhibitors, which include the JNK inhibitor SP600125, the p38 inhibitor SB203580, and the ERK inhibitor U0126, were obtained from Selleckchem (Houston, TX, USA). The ROS detection probes 2',7'-dichlorofluorescein diacetate (H2DCFDA) and MitoSOX were purchased from Abcam (Waltham, MA, USA) and Thermo Fisher Scientific (Waltham, MA, USA), respectively. Another ROS detection probe, dihydroethidium (DHE), was purchased from MilliporeSigma.

2.2. Animals

Male WT C57BL/6 mice (2 months old) were purchased from SLAC Laboratory Animal Co., Ltd. (Shanghai, China) and housed 5 per cage with free access to food and water in a common-temperature environment with a 12 h light–12 h dark cycle. All animal experiments were approved by the Experimental Animal Ethics Committee of Fujian Medical University (IACUC FJMU, 2023-Y-0360, March 3, 2023).

2.3. AAV-hTau expression

The AAV construct contains an expression cassette for the mCherry-hTau fusion gene (AAV-CMV-mCherry-hTau-WPRE) under the control of the CMV promoter (BrainVTA, Wuhan, Hubei, China). A recombinant AAV vector of serotype 9 was produced to express the longest tau isoform of hTau (4R/2N Tau). The recombinant AAV was delivered into the dorsal hippocampal cornu ammonis 3 (CA3) region at a rate of 125 nL/min via a microinjection system (Kd Scientific, Holliston, MA, USA) according to a previous study (Zhang et al., 2024b). The needle was kept in place for 5 min before withdrawal. Finally, the mice were sutured and placed on a heating pad for recovery. The mice were ready for experiments after one week.

2.4. Experimental design for the animal study

C57BL/6 mice were randomly assigned to four groups ($n = 10–12$ mice per group). The mice injected with AAV-hTau or AAV-vector were fed a normal diet with or without SSc diluted in 2% DMSO in PBS (0.0063% or 0.0125%, namely, 10 or 20 mg/kg/day) for 5 weeks (Fig. 1A). Behavioral tests were performed at the end of the SSc treatment. After the behavioral tests were completed, the mouse brain tissues were harvested for analysis of tau accumulation, tau phosphorylation, spine density, neuroinflammation, oxidative stress, superoxide dismutase 2 (SOD2) levels, and MAPK signaling.

2.5. Morris water maze (MWM) test

MWM was employed to analyze spatial learning and memory according to previous studies (Mei et al., 2025; Zhang et al., 2024b). A black circular tank (diameter 120 cm, depth 50 cm) was used to evaluate spatial learning and memory recall in the mice. The pool was divided into four quadrants, and a platform with a diameter of 9 cm was placed in one quadrant. The position of the platform was kept constant across the training. The pool was filled with water until the platform was 1 cm above the water surface in the visible platform test (day one) and 0.7 cm

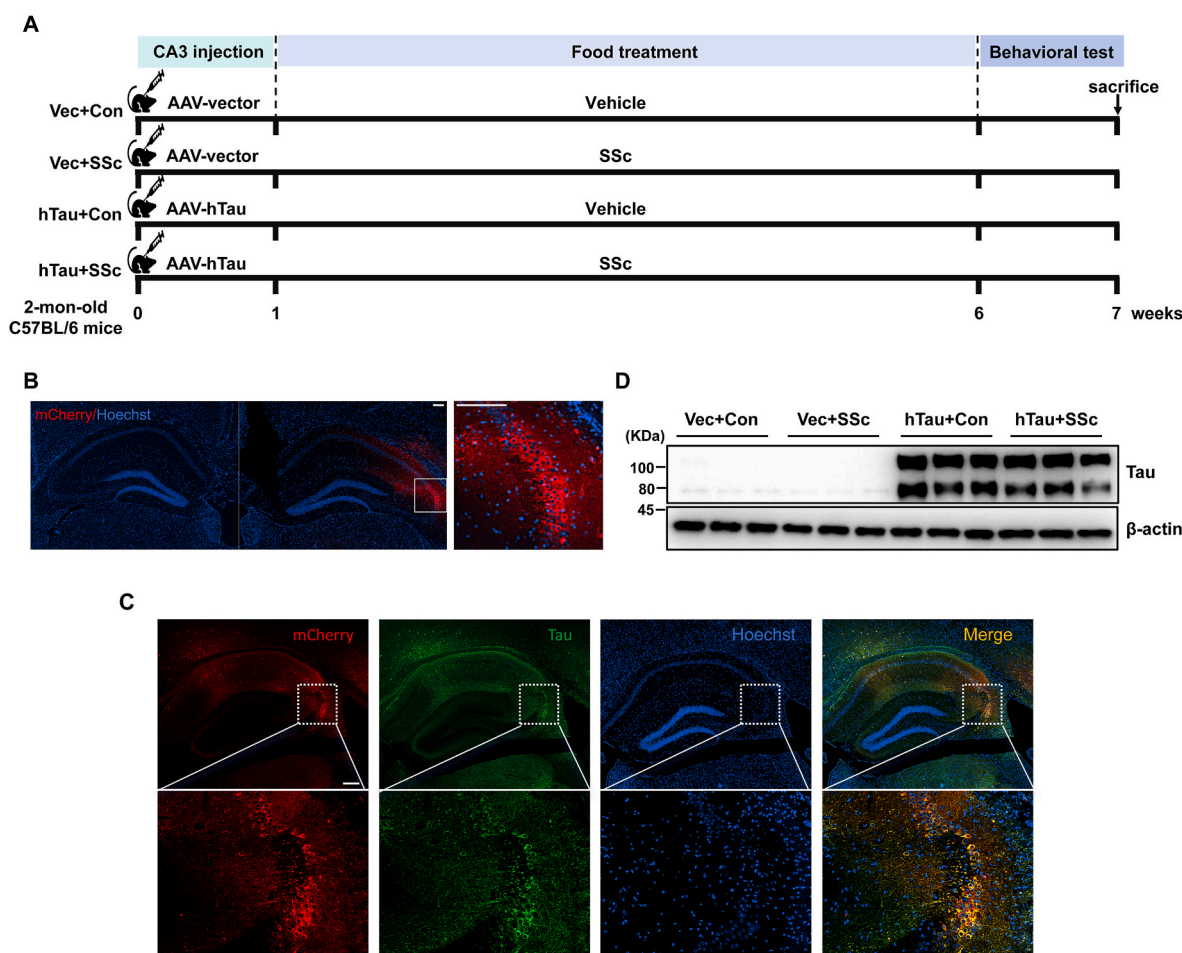


Fig. 1. The injection of AAV-hTau successfully induces tau expression in the brains of WT mice. (A) Overview of the experimental scheme. At age 2 months, all the mice were randomly assigned to four groups ($n = 10–12$ mice per group) (AAV-vector vehicle, AAV-vector SSc, AAV-hTau vehicle, and AAV-hTau SSc). All groups of mice were subjected to stereotaxic brain injection, food treatment, behavioral tests, and sacrifice for histological and biochemical analyses, as described. (B) Representative images of viral expression in right hippocampal sections one week after the injection. hTau was labeled with mCherry (red), and nuclei were labeled with Hoechst 33342 (blue). The scale bar represents 200 μm ; the inset scale bar represents 200 μm . (C) Representative images showing the co-localization of mCherry (red) and hTau (green) of viral expression in right hippocampal sections six weeks after the injection. Nuclei were labeled with Hoechst 33342 (blue). The scale bar represents 200 μm ; the inset scale bar represents 50 μm . (D) Confirmation of hTau expression in mouse brains via the hTau-specific monoclonal antibody HT7 ($n = 3$ per group). β -actin expression was used as a loading control.

below the water surface in the hidden platform test (days two to five). A flag was placed on the platform to increase its visibility only on day one. The water was made opaque with nontoxic white powder to hide the platform from sight. The temperature of the water was maintained at 24 °C.

The mice were transferred to the behavioral testing room and acclimated for at least 1 h before the test. The mice were trained to find the platform for five consecutive days, with five trials per day. The ability of the mice to locate the platform was recorded with a camera (RWD, Shenzhen, China). In the training sessions (days one to five), the mice were allowed to swim for 60 s. If the latency to find the platform was less than 60 s, the mice were guided to the platform and remained there for 5 s. Otherwise, the mice were guided to the platform and remained there for 20 s. Each subsequent trial began with a different starting direction. On day six (24 h after training), the hidden platform was removed, and the mice were subjected to the probe trial test for 60 s. In each trial, the mice were gently placed into the water with their heads facing the quadrant at the edge of the pool. All MWM tests were carried out at the same time every day in the afternoon. The escape latency, time spent in each quadrant and speed of swimming were analyzed via SMART video tracking software (version 3.0.06) provided by the Panlab Harvard Apparatus (Barcelona, Spain).

2.6. Tissue preparation

After the behavioral tests, the randomly selected mice were deeply anesthetized with 1.25% tribromoethanol (MilliporeSigma) and perfused transcardially with 0.9% NaCl solution. The brains of the mice were rapidly removed, and the whole brain tissue was immediately transferred to neutral buffered paraformaldehyde (4% in PBS) for fixation, embedded in paraffin, and then cut into serial coronal sections (3–5 µm), which were stored at 4 °C for immunohistochemical analysis and immunofluorescence analysis. The hippocampal tissues from the right hemisphere were stored at –80 °C for Western blotting.

2.7. Histological analysis

For immunofluorescence, tissue sections were blocked with goat serum (Boster Bio, Wuhan, Hubei, China) at room temperature for 1 h after deparaffinization, rehydration and antigen retrieval. Afterward, the tissue sections were incubated with primary antibodies in PBS containing 10% goat serum at 4 °C overnight (Supplementary Table S1). The next day, the slides were incubated with Alexa Fluor 488- or 546-conjugated secondary antibodies (Invitrogen, Carlsbad, CA, USA) for 1 h at room temperature in the dark. Nuclei were counterstained with Hoechst 33342 for 5 min (MilliporeSigma) and covered with antifade mounting medium (Southern Biotech, Birmingham, AL, USA). Images were acquired with both a conventional fluorescence microscope (Zeiss Axio Imager 2, Oberkochen, Germany) and a confocal microscope (Leica SP5, Wetzlar, Germany).

For immunohistochemistry, tissue sections were treated with 3% H₂O₂ solution at room temperature for 10 min to suppress endogenous peroxidase activity after deparaffinization, rehydration and antigen retrieval. The tissue sections were then blocked with goat serum at room temperature for 1 h and incubated with the primary antibody in PBS containing 10% goat serum at 4 °C overnight (Supplementary Table S1). Afterward, the tissue sections were incubated with horseradish peroxidase (HRP)-conjugated secondary antibodies (Boster Bio) for 1 h at room temperature in the dark. The sections were processed with DAB reagents via a DAB kit (MXB Biotechnologies, Fuzhou, Fujian, China). The sections were subsequently counterstained with hematoxylin and visualized via a digital microscope (REB-01, ECHO, San Diego, CA, USA).

2.8. Golgi/Cox staining

Golgi/Cox staining was conducted via the FD Rapid GolgiStain™ Kit

(FD NeuroTechnologies, Inc., Columbia, MD, USA) following the manufacturer's instructions. The brains were immersed in impregnation solution for fixation and cut into sections (200 µm) via a vibratome (Leica, Wetzlar, Germany). The sections were mounted on gelatin-coated microscope slides and stained. Finally, the dendritic spines were imaged with a Zeiss Primo Star microscope with a 100× oil objective and quantified with ImageJ software (version 1.51j8).

2.9. RNA sequencing profiling and differentially expressed gene (DEG) analysis

RNA sequencing profiling was performed according to our previous report (Li et al., 2023). Total RNA was extracted from whole hippocampal tissues via TRIzol. The RNA concentration and purity were assessed with a Nanodrop 2000 spectrophotometer (Thermo Fisher Scientific), and the RNA integrity was quantified with an Agilent 5300 system (Agilent Technologies, Santa Clara, CA, USA). The RNA quality number (RQN) for the samples was greater than 6.5. RNA sequencing was performed on an Illumina NovaSeq 6000 system (Shanghai Majorbio Biopharm Biotechnology Co., Ltd., Shanghai, China). The RNA-seq data shown in this study have been deposited in the Genome Sequence Archive (GSA) and are accessible through the GSA series accession number CRA025339.

Gene classification was performed via the DAVID tool (<http://david.abcc.ncifcrf.gov/>) and the Medline database (<http://www.ncbi.nlm.nih.gov/>). A clustering heatmap of RNA expression was generated via MeV software (<https://webmev.tn4.org>) via the correlation of normalized fragments per kilobase of exon per million reads (FPKM) values across samples. For the gene set enrichment analysis (GSEA), we used the GSEA algorithm and software developed by the Broad Institute (Liberzon et al., 2011; Subramanian et al., 2005). A functional map was produced through an enrichment map (Merico et al., 2010) in Cytoscape (Shannon et al., 2003).

2.10. Cell culture

Human embryonic kidney (HEK293) cells and human neuroblastoma SH-SY5Y and SK-N-SH cells were obtained from the American Type Culture Collection (ATCC) (Manassas, VA, USA). HEK293 cells, HEK293/hTau cells, which stably overexpress the longest form of WT hTau (Liu et al., 2012; Wang et al., 2022b), and SK-N-SH cells were cultured in Eagle's minimum essential medium (EMEM) supplemented with 10% fetal bovine serum (FBS) at 37 °C in a humidified atmosphere containing 5% CO₂. SH-SY5Y cells were cultured in medium containing a 1:1 ratio of EMEM and Ham's F-12 nutrient mixture (HyClone, Logan, UT, USA) supplemented with 10% FBS, supplemented with 1% penicillin/streptomycin at 37 °C in a humidified atmosphere containing 5% CO₂.

2.11. Cytotoxicity assay

After treatment with AlCl₃, the viability of HEK293, HEK293/hTau, SK-N-SH and SH-SY5Y cells was assessed via MTT and trypan blue dye exclusion assays. Cellular metabolic activity was evaluated via a colorimetric MTT assay (thiazolyl blue tetrazolium bromide; MilliporeSigma) in accordance with the manufacturer's protocols. The cells were incubated with the MTT labeling agent (0.5 mg/mL) for 3 h. The supernatant was removed, and the purple MTT-formazan crystals formed were dissolved in DMSO. The optical density (OD) values were assessed and recorded at 570 nm absorbance via a microplate reader (SpectraMax i3, Molecular Devices, Sunnyvale, CA, USA). Overall cell viability determined via trypan blue staining was calculated as the number of unstained viable cells divided by the total number of cells.

2.12. Measurement of the cellular ROS content

Following the treatment of HEK293, HEK293/hTau, SK-N-SH and SH-SY5Y cells with AlCl₃ in the absence or presence of SSc, the ROS levels were measured by using three different fluorescent probes. Cytosolic H₂O₂ levels were assessed by incubating the cells with H2DCFDA (1 μM; for 40 min) in the dark at 37 °C. Cytosolic and mitochondrial superoxide levels were assessed by incubating the cells with DHE (20 μM; for 60 min) and MitoSOX (1 μM; for 60 min) in the dark at 37 °C. The fluorescence was visualized under a fluorescence microscope (Nikon ECLIPSE TE2000). The green fluorescence intensity of H2DCFDA and the red fluorescence intensities of DHE and MitoSOX were quantified via ImageJ.

2.13. Western blot analysis

Western blot analysis was carried out as described previously (Gan et al., 2022; Wang et al., 2022a; Zhang et al., 2022, 2023). Briefly, protein was extracted from mouse brain and mammalian cells via RIPA buffer (Beyotime Biotechnology Co., Ltd., Shanghai, China) supplemented with protease and phosphatase inhibitors. The same amounts of proteins were electrophoretically separated on a sodium dodecyl sulfate (SDS)-polyacrylamide gel followed by transfer onto polyvinylidene fluoride (PVDF) membranes. After the membranes were blocked with 5% milk-TBST (Tris-buffered saline, 0.1% Tween 20) or 5% BSA-TBST at room temperature for 1 h, the proteins were incubated with the following primary antibodies overnight at 4 °C (Supplementary Table S1). The proteins were then incubated with horseradish peroxidase-conjugated anti-mouse or anti-rabbit secondary antibodies. The immunoreactive bands were visualized via a chemiluminescence Western blot detection system (WESTAR, Cyanagen, Bologna, Italy). For the capture and quantitative analysis of the Western blots, a Fusion Solo imaging system (Vilber, Lourmat, France) was used.

2.14. Statistical analysis

Statistical analysis was performed via GraphPad Prism software (version 8.3.0). The values are presented as the means ± standard errors of the means (SEMs) obtained from triplicate experiments. The *P* values were calculated via analysis of variance (ANOVA) followed by Dunnett's or Sidak's multiple comparison *post hoc* test, and the significance levels were set at *P* < 0.05, *P* < 0.01, and *P* < 0.001.

3. Results

3.1. Expression of hTau in the mouse brain after AAV-hTau injection

A summary of the experimental design is depicted in Fig. 1A. Stereotaxic injections of AAV-vector or AAV-hTau were performed in the brains of 2-month-old C57BL/6 mice. After 1 week, the mice in both the AAV-vector group and the AAV-hTau group were orally administered either saline or SSc via *ad libitum* access to food and water for 5 weeks. It has been demonstrated the presence of SSc in brain microdialysate following administration, indicating that SSc is capable of crossing the blood-brain barrier (BBB) in rats after oral gavage with Chaigui granule, which contains *Radix Bupleuri* (Gao et al., 2020). Furthermore, we evaluated the ability of SSc to penetrate the BBB using *in silico* analysis of its absorption, distribution, metabolism, excretion, and toxicity (ADMET) profiles via ADMETlab 3.0 (Fu et al., 2024). The results indicate that SSc possesses the capacity to cross the BBB. Following this treatment period, all the mice were subjected to behavioral tests. The mCherry-hTau fusion protein was clearly transferred to and expressed in the mouse hippocampal CA3 region of the right hemisphere one week (Fig. 1B) and six weeks after the injection (Fig. 1C). AAV-mediated overexpression of hTau in the mouse hippocampus was also observed via Western blotting with an hTau-specific monoclonal antibody (clone

HT7) (Fig. 1D).

3.2. Oral administration of SSc alleviates cognitive impairments in mice

Before the behavioral tests, we assessed the *in vivo* toxicity of SSc and did not observe changes in the mortality or body weight of the mice receiving SSc treatment either alone or in combination with the AAV constructs (Supplementary Fig. S2A and C). Similarly, we found no adverse histopathological features, such as inflammation, in any of the tissues evaluated after each treatment (Supplementary Fig. S2B). In addition to these observations, no other obvious general pathological signs or changes were found in any of the SSc-treated mouse groups throughout the monitoring period. We employed the MWM test to determine the effects of SSc on learning and memory deficits in hTau mice. During the training session, hTau expression in the brain increased the latency of the mice, suggesting that hTau overexpression impaired cognition. However, the escape latency and average escape latency of the hTau mice fed the SSc diet were much shorter than those of the hTau mice fed the normal diet on days five and days two to five, respectively (Fig. 2A–C).

SSc-mediated improvements in spatial learning and memory were also observed, with increased time spent in the target quadrant (Fig. 2D and E). The hTau-injected mice spent less time in the target quadrant than did the AAV vehicle-injected mice. Notably, compared with the normal diet, the SSc diet significantly enhanced learning ability, as indicated by the hTau-injected mice spending a greater percentage of time in the target quadrant. The swimming speed was similar among the treatment groups (Fig. 2F). Thus, a decrease in the escape latency to find the platform and an increase in the time spent in the target quadrant in the MWM test indicated that SSc treatment alleviated memory deficits in the mouse model of tau pathology.

3.3. SSc attenuates the abnormal phosphorylation of tau in the brains of AAV-hTau-injected mice

The abnormal phosphorylation of tau is significantly increased in AD and other tauopathies and leads to neurofibrillary degeneration in AD (Bramblett et al., 1993; Buerger et al., 2002; Goedert et al., 1992; Iqbal et al., 2005). Tau phosphorylation at the Thr231 residue, a potential biomarker of neurofibrillary pathology in AD and tauopathy, was substantially increased in the AAV-hTau group, and SSc treatment at a 20 mg/kg/day dose strongly alleviated this increase in tau phosphorylation (Fig. 3A and B), whereas the 10 mg/kg/day dose of SSc moderately attenuated tau hyperphosphorylation (Supplementary Fig. S3). Western blotting performed via phosphorylation site-specific tau antibodies also revealed the overt and extensive phosphorylation of tau at the Ser202/Thr205, Thr231, and Ser396 residues in the mice with tauopathy caused by AAV-mediated hTau transfer, and SSc treatment markedly reduced the levels of phosphorylated tau (Fig. 3C and D).

3.4. SSc ameliorates neurogenic and synaptic deficits in the AAV-hTau mouse brain

Tauopathies, such as AD, are characterized by decreased adult hippocampal neurogenesis (Mu and Gage, 2011). Doublecortin (DCX), a marker of neurogenesis, has been suggested to regulate dendritic spine formation through its microtubule-binding activity (Cohen et al., 2008; Francis et al., 1999). DCX expression in the hippocampal dentate gyrus (DG) region, as determined by immunohistochemistry, was apparently reduced in the AAV-hTau group, and this reduction was significantly attenuated by SSc treatment (Fig. 4A and B, and Supplementary Fig. S4). Spine density in the hippocampus, as a correlate of excitatory synaptic strength, was estimated in the mid-apical and basal dendrites of CA1 pyramidal neurons via Golgi/Cox staining. Consistent with the results of DCX expression, a significant reduction in the dendritic spine density in the CA1 region was observed in the AAV-hTau-injected mice compared

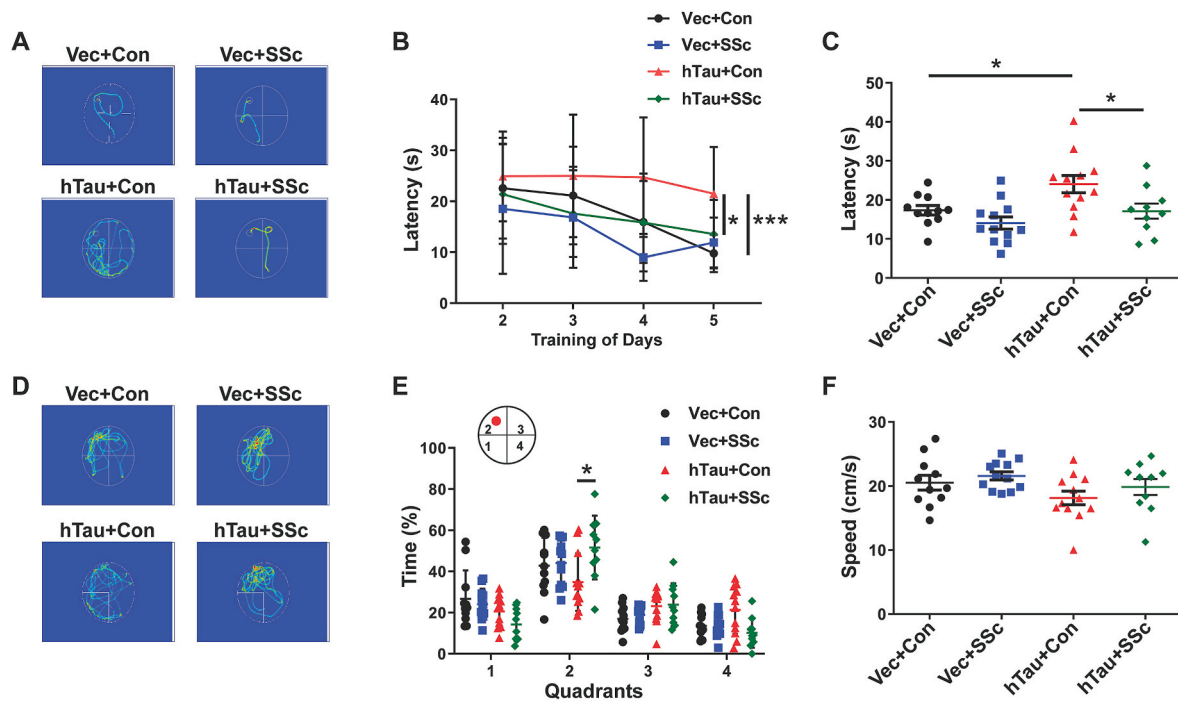


Fig. 2. SSc exerts a cognitive-promoting effect on memory impairment in AAV-hTau mice. The MWM test was used to assess the cognitive function of AAV-hTau mice treated with SSc (20 mg/kg/day). (A) Representative track plots of the MWM test during the training trials for the mice ($n = 10$ – 12 per group) obtained via video-tracking software (Smart v3.0). (B) Acquisition task: latency to find the (hidden) platform during training days. $*P < 0.05$ and $***P < 0.001$; two-way ANOVA/Dunnett's test. (C) Average latency to find the (hidden) platform during the training days. $*P < 0.05$; one-way ANOVA/Dunnett's test. (D) Representative occupation plots during the probe trial. (E) Retention task: percentage of time spent in the target quadrant or nontarget quadrants during the probe test. $*P < 0.05$; one-way ANOVA followed by Dunnett's test. (F) Mean swimming speed to find the platform during periods of active swimming over 5 days.

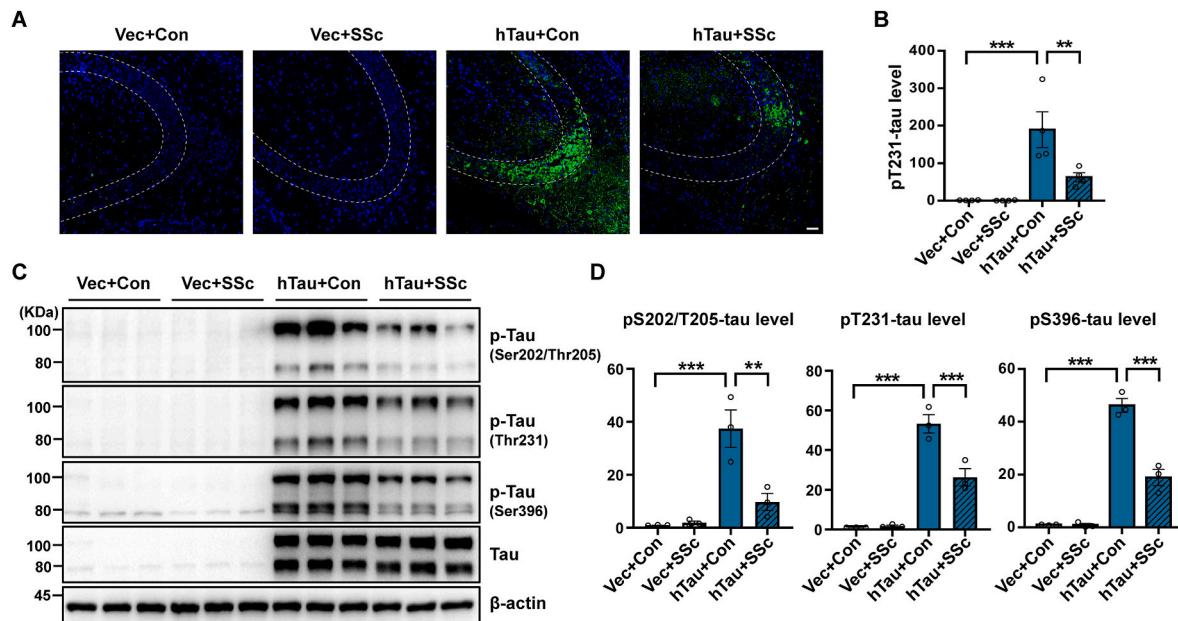


Fig. 3. SSc treatment significantly attenuates tau phosphorylation in the AAV-hTau mouse brain. Immunohistochemical staining and Western blotting were performed to assess tau phosphorylation in AAV-hTau mice treated with SSc (20 mg/kg/day). (A) Representative images of immunohistochemical staining of coronal mouse brain sections from each treatment group stained with a phospho-Thr231-tau (green) antibody. The cell nuclei were stained with Hoechst 33342 (blue). The scale bar represents 50 μ m ($n = 4$ per group). (B) Quantification of the immunofluorescence intensity of phospho-Thr231-tau in the hippocampus. $**P < 0.01$ and $***P < 0.001$; one-way ANOVA/Dunnett's test. (C) Expression levels of total hTau (HT7 antibody) and site-specific tau phosphorylation in the mouse brain ($n = 3$ per group). β -actin was included as a loading control. (D) Quantification of tau phosphorylation at the Ser202/Thr205, Thr231 and Ser396 sites. The data were normalized to the levels of β -actin. $**P < 0.01$ and $***P < 0.001$; one-way ANOVA/Dunnett's test.

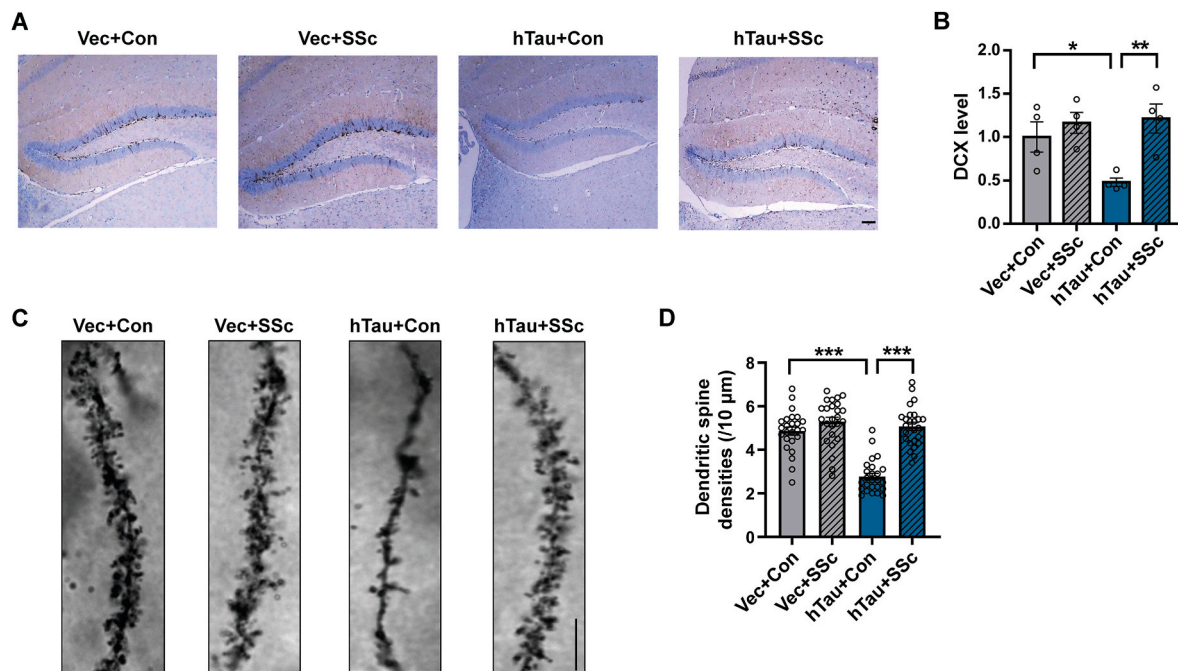


Fig. 4. SSc treatment attenuates deficits in dendritic spine density in the hippocampus in AAV-hTau-injected mice. Immunohistochemical staining and Golgi staining were carried out to assess the attenuation of synaptic deficits in AAV-hTau mice treated with SSc (20 mg/kg/day). (A) Representative images of immunohistochemical staining for DCX in the hippocampal DG region of each treatment group. The scale bar represents 50 μm (n = 4 per group). (B) Quantification of DCX immunoreactivity. * $P < 0.05$ and ** $P < 0.01$; one-way ANOVA/Dunnett's test. (C) Representative images of Golgi staining showing the dendritic spine density in the hippocampal CA3 region of the mouse brain (n = 5 per group). The scale bar represents 50 μm. (D) Quantification of the dendritic spine density as spines per 10 μm (n = 5 images per mouse). *** $P < 0.001$; one-way ANOVA/Dunnett's test.

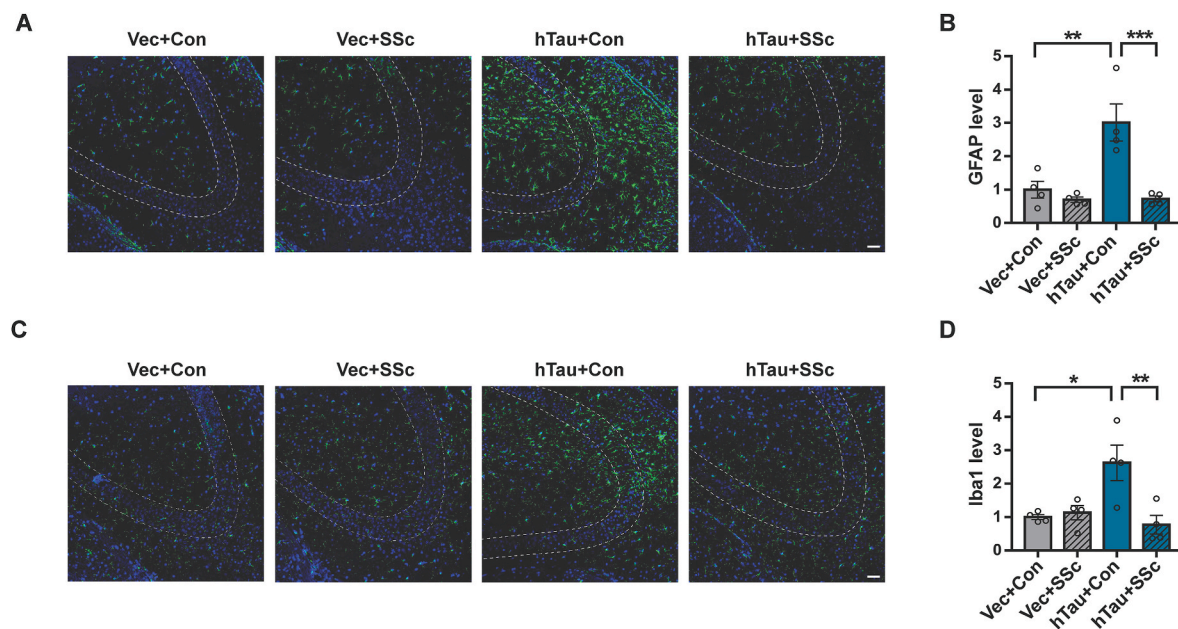


Fig. 5. SSc treatment significantly alleviates tauopathy-related neuroinflammation in mice. Immunohistochemical staining was performed to assess neuroinflammation in AAV-hTau mice treated with SSc (20 mg/kg/day). (A) Representative images of immunohistochemical staining of coronal mouse brain sections from each treatment group stained with a GFAP (green) antibody. The cell nuclei were stained with Hoechst 33342 (blue). The scale bar represents 50 μm (n = 4 per group). (B) Quantification of the immunofluorescence intensity of GFAP in the hippocampus. ** $P < 0.01$, *** $P < 0.001$; one-way ANOVA/Dunnett's test. (C) Representative images of immunohistochemical staining of coronal mouse brain sections from each treatment group stained with an Iba1 (green) antibody. The cell nuclei were stained with Hoechst 33342 (blue). The scale bar represents 50 μm (n = 4 per group). (D) Quantification of the immunofluorescence intensity of Iba1 in the hippocampus. * $P < 0.05$, ** $P < 0.01$; one-way ANOVA/Dunnett's test.

with the AAV-vector-injected mice of the same age in the mid-apical dendritic segments of pyramidal cells, and this phenomenon was strongly relieved by SSc (Fig. 4C and D).

3.5. SSc decreases neuroinflammation in the AAV-hTau mouse brain

Neuroinflammation associated with astrocytes and microglia is a driving force in AD (Heneka et al., 2015). Given that astrocytes and microglia are identified by GFAP+ and Iba1+ immunoreactivity, respectively (Ahmed et al., 2007; Takizawa et al., 2008), and that both GFAP and Iba1 are neuroinflammatory markers (Norden et al., 2016), immunofluorescence staining was conducted with GFAP and Iba1 antibodies. The levels of GFAP and Iba1 were elevated in the AAV-hTau-injected mice. SSc treatment decreased the numbers of GFAP + astrocytes and Iba1 + microglia in the AAV-hTau-injected mice (Fig. 5 and Supplementary Fig. S5), indicating that the treatment of tauopathy model mice with SSc reduced neuroinflammation.

3.6. Transcriptome profiling and functional enrichment analysis of AAV-hTau-injected mice with SSc

RNA-Seq was used to characterize the mechanism underlying the therapeutic effects of SSc on tauopathy in mouse brain samples. Hierarchical clustering analysis and a heatmap of the DEGs identified in the hTau_SSc group compared with the hTau_con group revealed that 3310 genes whose expression levels were significantly altered were clustered

(P value < 0.05); among them, 3172 genes whose expression was upregulated and 138 whose expression was downregulated in the hTau_SSc group compared with the hTau_con group. In contrast, 385 genes were upregulated and 2925 were downregulated in the hTau_con group compared with the Vec_con group (Fig. 6A).

In addition, GSEA of the HALLMARK gene set and GOBP gene set revealed several canonical pathways that were significantly altered due to AAV-hTau injection or SSc treatment (Fig. 6B and C). For each gene set, the threshold for GSEA was set at a | normalized enrichment score (NES) | > 1 and a nominal P value < 0.05. In the comparison of the hTau_con group and the Vec_con group, among the top 5 GSEA-enriched HALLMARK gene sets and GOBP gene sets in Fig. 6B, gene enrichment plots showing biological processes such as oxidative phosphorylation, PI3K-Akt-mTOR signaling, dendritic spine development, and the regulation of dendritic spine development are depicted in Fig. 6D. In the comparison of the hTau_SSc group and the hTau_con group, among the top 5 GSEA-enriched HALLMARK gene sets and GOBP gene sets in Fig. 6C, gene enrichment plots showing biological processes such as oxidative phosphorylation, hypoxia, the neuropeptide signaling pathway, and dendritic spine development are depicted in Fig. 6E. Visualization of the GSEA results via Cytoscape revealed that most of the DEGs were involved in the MAPK pathway, Rho pathway, oxidative phosphorylation, and neurotransmitter regulation (Fig. 6F). These results indicate that oxidative stress and the MAPK pathway may be associated with the molecular mechanisms underlying tauopathy induced by AAV-mediated hTau gene transfer and the therapeutic

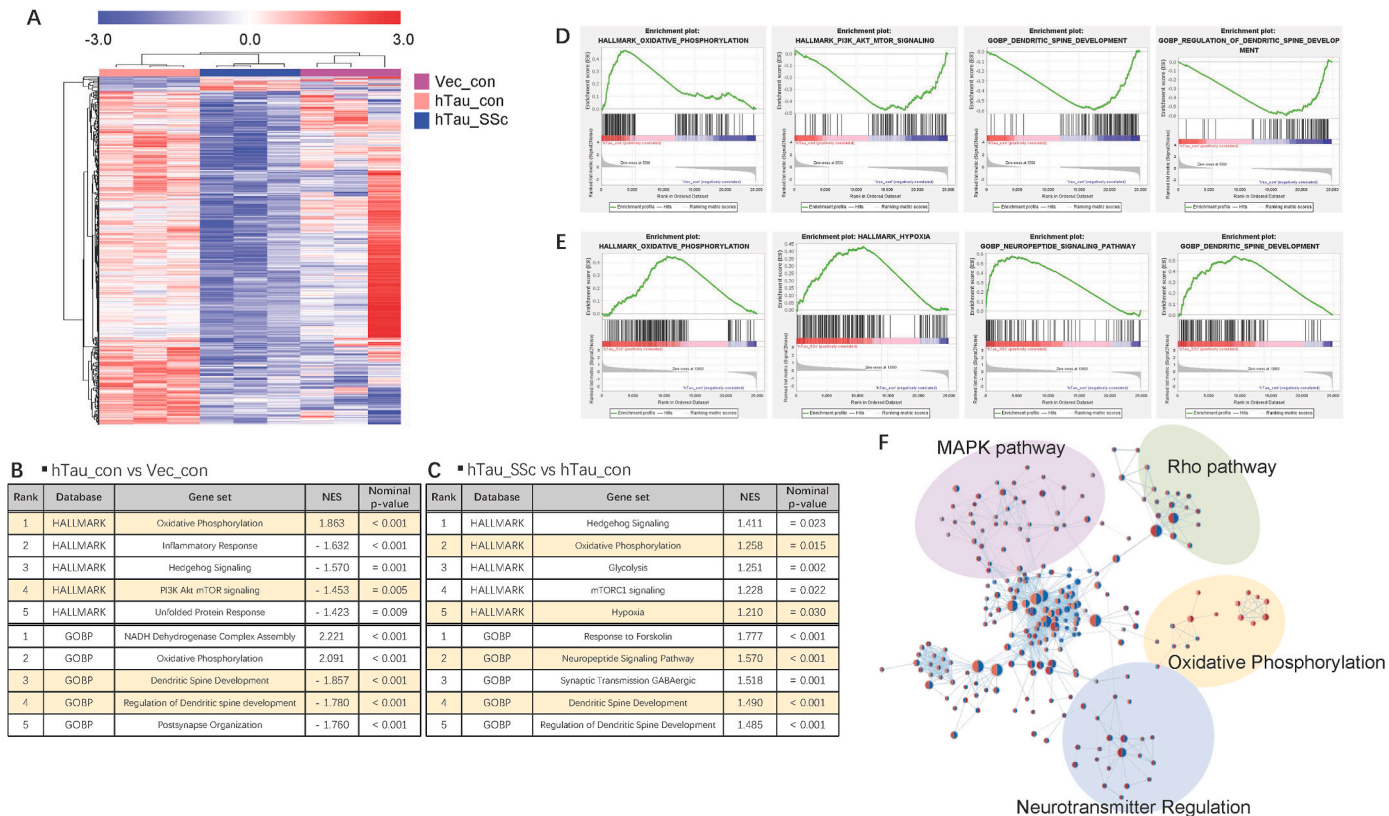


Fig. 6. Transcriptomic analysis of AAV-hTau mice with SSc. (A) Hierarchical clustering analysis and heatmap of 3310 DEGs between the hTau_SSc and hTau_con groups (fold change >1, P value < 0.05). Both rows and columns are clustered via the Pearson correlation distance and average linkage. (B) The top 5 enriched pathways in the HALLMARK and GOBP databases are annotated for the comparison of the hTau_con and Vec_con groups. (C) The top 5 enriched pathways in the HALLMARK and GOBP databases are annotated for the comparison of the hTau_SSc and hTau_con groups. (D) Top four enrichment plots of the GSEA results for the comparison of the hTau_con and Vec_con groups from B. (E) Top four enrichment plots of the GSEA results for the comparison of the hTau_SSc and hTau_con groups from C. (F) Pathway analysis visualizing the GSEA data via enrichment map software. Each node corresponds to a gene set from the GSEA data. The node size represents the number of genes in the pathway gene set, and the edges connect pathways with similar gene sets. The right hemisphere of the node is from the 'hTau_con vs. Vec_con' comparison, and the left hemisphere is from the 'hTau_SSc vs. hTau_con' comparison. Red indicates upregulation, and blue indicates downregulation.

potential of SSc.

3.7. SSc alleviates oxidative stress in the AAV-hTau mouse brain

Oxidative stress is a pathological characteristic of AD and other tauopathies and is known to contribute to tau hyperphosphorylation as an upstream event related to tau pathology (Alavi Naini and Soussi-Yanicostas, 2015). Based on the GSEA of the RNA-Seq data, we evaluated the oxidative stress-related biomarkers dityrosine (DT) and SOD2 in the hippocampal regions of the brain to evaluate the possible antioxidant activity of SSc *in vivo*. Immunostaining with antibodies against DT and SOD2 revealed an increase in DT levels and a decrease in SOD2 levels in the brains of the AAV-hTau-injected mice compared with those in the brains of the AAV-Vec-injected mice. These adverse effects could be relieved by SSc treatment (Fig. 7 and Supplementary Fig. S6).

3.8. SSc reduces MAPK phosphorylation in the brains of AAV-hTau-injected mice

MAPKs play crucial roles in oxidative stress-mediated cytotoxicity and survival. JNK and p38 MAPK, members of the MAPK family, are activated by various stresses and are key regulators of stress-activated diseases, including AD (Bendotti et al., 2006; Munoz and Ammit, 2010; Savage et al., 2002; Yarla et al., 2016). The abnormal phosphorylation and activation of ERK, another member of the MAPK family, are also linked to oxidative stress-mediated AD. Based on the GSEA of RNA-Seq data, we analyzed the activation (estimated phosphorylation status) of MAPK family members in the hippocampal regions of the brain to evaluate the potential of targeting MAPK via SSc treatment *in vivo*. Western blotting with phospho-MAPK antibodies revealed that the phosphorylation and activation of JNK, p38, and ERK were increased in the brains of the AAV-hTau-injected mice and that, notably, SSc decreased the phosphorylation and activation of these MAPKs (Fig. 8).

3.9. SSc protects against AlCl₃-induced oxidative stress and attenuates MAPK activation and tau hyperphosphorylation in cells

To further investigate the possible antioxidant activity of SSc in terms of tauopathy, we introduced AlCl₃, which is known to induce tau pathology and oxidative stress-induced cytotoxicity (Walton, 2007; Yu et al., 2014; Zhao et al., 2020). As expected, compared with HEK293 cells, HEK293/hTau cells were more susceptible to AlCl₃-induced cytotoxicity because of the high expression of tau. SSc exerted protective effects on AlCl₃-induced cytotoxicity in both HEK293 and HEK293/hTau cells, and interestingly, the protective effect of SSc was more prominent in HEK293/hTau cells (Fig. 9A and C). These findings suggest that the induced tau protein might be a target protein involved in the protective effects of SSc on AlCl₃-induced cytotoxicity. SSc also exerted protective effects on AlCl₃-induced cytotoxicity in neuroblastoma SK-N-SH and SH-SY5Y cells (Fig. 9B and D), both of which express the endogenous tau protein.

AlCl₃ can induce oxidative damage, which mediates neurotoxicity in a rat model of AD (Dey and Singh, 2022; Skalny et al., 2021; Zhao et al., 2020). As shown in Fig. 9E–J, AlCl₃ triggered the generation of various ROS, such as cytosolic H₂O₂ (assessed by H2DCFDA), cytosolic superoxide (assessed by DHE), and mitochondrial superoxide (assessed by MitoSOX), in both HEK293 and HEK293/hTau cells. Notably, SSc significantly reduced the levels of these ROS in the treated cells. In particular, SSc-mediated antioxidant activities against cytosolic H₂O₂ and mitochondrial superoxide were more prominent in HEK293/hTau cells (Fig. 9E, F, I and J), indicating that the SSc-mediated protective effects on AlCl₃-induced oxidative stress occur in a tau-dependent manner. Moreover, AlCl₃-triggered ROS generation in SK-N-SH and SH-SY5Y cells was significantly attenuated by SSc pretreatment (Supplementary Fig. S7).

Time-course experiments revealed that AlCl₃ first activated MAPK signaling (ERK, JNK, and p38), followed by phosphorylation of tau in

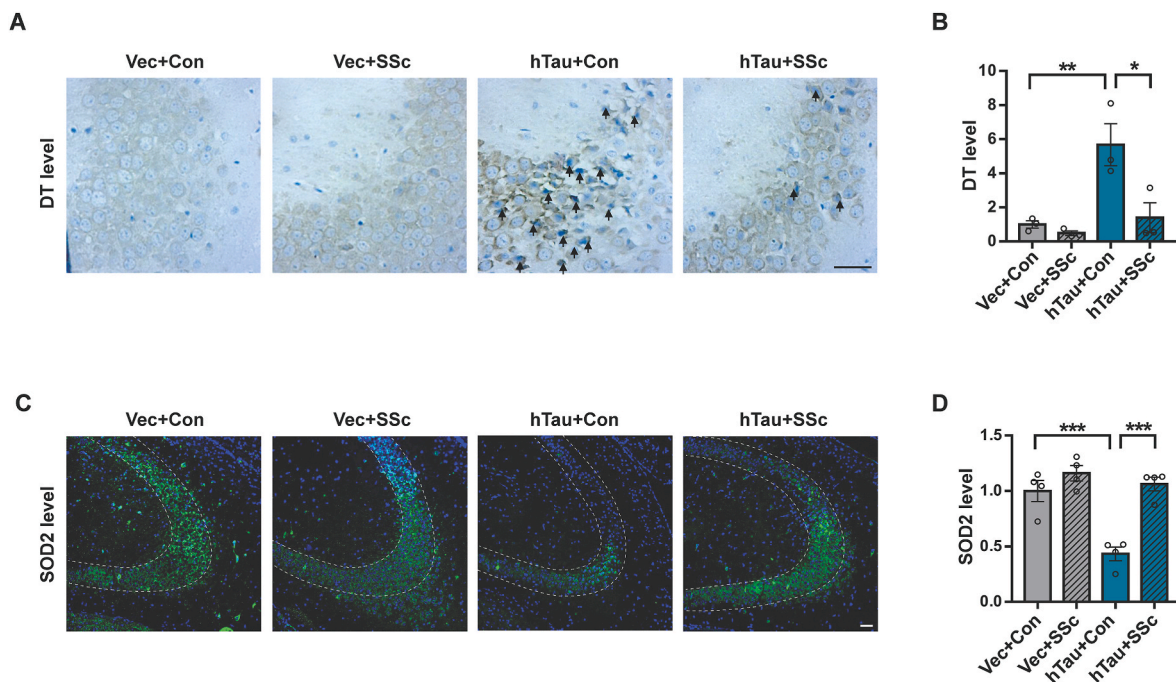


Fig. 7. SSc treatment significantly attenuates oxidative stress in AAV-hTau mice. Immunohistochemical staining was used to assess oxidative stress in AAV-hTau mice treated with SSc (20 mg/kg/day). (A) Representative images of immunohistochemical staining of coronal mouse brain sections from each treatment group with a mouse monoclonal DT antibody. The scale bar represents 50 μ m ($n = 3$ per group). (B) Quantification of the immunofluorescence intensity of DT in the CA3 region of the hippocampus by immunohistochemistry via ImageJ software. $*P < 0.05$ and $**P < 0.01$; one-way ANOVA/Dunnett's test. (C) Representative images of immunohistochemical staining of coronal mouse brain sections from each treatment group stained with a SOD2 (green) antibody. The cell nuclei were stained with Hoechst 33342 (blue). The scale bar represents 50 μ m ($n = 4$ per group). (D) Quantification of the immunofluorescence intensity of SOD2 in the CA3 region of the hippocampus. $***P < 0.001$; one-way ANOVA/Dunnett's test.

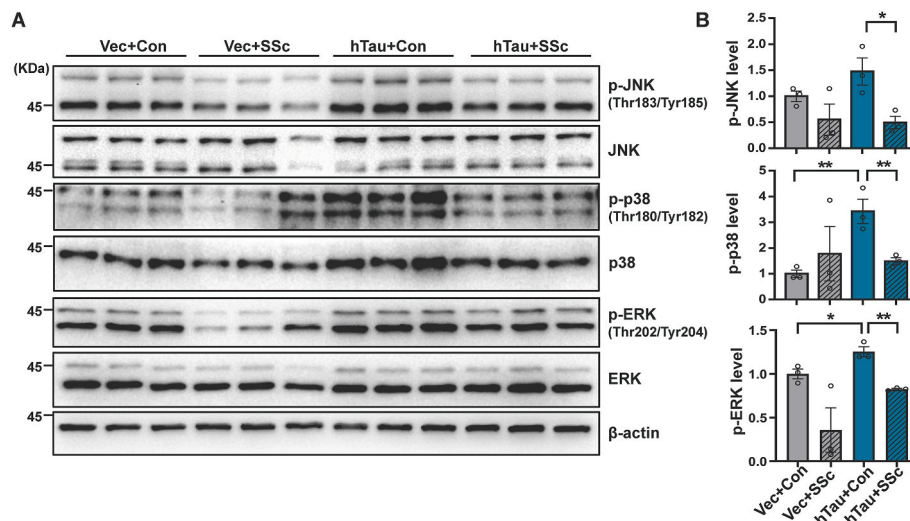


Fig. 8. The tauopathy-related phosphorylation of MAPK is reduced by SSc. Western blotting was performed to assess the phosphorylation of MAPK in AAV-hTau mice treated with SSc (20 mg/kg/day). (A) Western blot analysis of the levels of JNK phosphorylated at Thr183/Tyr185, p38 phosphorylated at Thr180/Tyr182, ERK1/2 phosphorylated at Thr202/Tyr204 and total protein in the brain lysates ($n = 3$ per treatment group). (B) Bar graphs for quantitative comparison via densitometry. * $P < 0.05$ and ** $P < 0.01$; one-way ANOVA/Dunnett's test.

HEK293/hTau cells (Supplementary Fig. S8). Treatment with ROS scavengers (NAC for H_2O_2 and tempol for superoxide) significantly reduced both MAPK activation and tau hyperphosphorylation (Supplementary Fig. S9), indicating that ROS production mediates MAPK-driven tau phosphorylation. Furthermore, pharmacological inhibition of JNK (SP600125), p38 (SB203580), or ERK (U0126) reduced tau phosphorylation at AD-relevant sites (Supplementary Fig. S10), confirming MAPK involvement in tau pathology. Consistent with these findings, SSc treatment inhibited $AlCl_3$ -induced phosphorylation of MAPKs and tau in a time-dependent manner in HEK293/hTau cells (Fig. 10). These results suggest that SSc reduces $AlCl_3$ -induced oxidative stress, thereby suppressing ROS-dependent MAPK activation and downstream tau hyperphosphorylation. Collectively, our data demonstrate that SSc confers neuroprotection in tauopathy-related cellular models by mitigating oxidative stress and interfering with the MAPK signaling pathway, ultimately attenuating tau hyperphosphorylation.

4. Discussion

Previously, we reported that SSc inhibits abnormal tau phosphorylation and the amyloidogenic processing of amyloid precursor protein (APP) in a neuronal cell culture model (Lee et al., 2016), both of which are major factors contributing to AD. In particular, abnormal phosphorylation of the neuronal tau protein leads to the formation of aggregated forms such as filaments and neurofibrillary tangles, and these aggregates are implicated in the pathogenesis of tauopathies, including AD (Goedert et al., 1992; Iqbal et al., 2005). Herein, we report the *in vivo* therapeutic potential of SSc for treating neurodegenerative tauopathies in a mouse model of tauopathy. We administered a tau pathology-inducing AAV expressing human full-length WT 4R/2N tau into the brains of mice to generate an engineered mouse model of tauopathy. We also report the therapeutic mechanisms of SSc as a disease-modifying therapy to combat tauopathies.

Since the overexpression of human full-length tau in WT mice is known to induce memory deficits along with impaired synaptic plasticity (Yin et al., 2016), we performed an MWM test using mice expressing AAV-hTau. Pilot experiments involving the administration of SSc via oral gavage revealed that supplementation with SSc improved learning and long-term memory in mice (data not shown). Based on a preliminary study, for promising clinical application, we provided food containing SSc to the mice daily. The MWM test results revealed that SSc

alleviated the cognitive impairment of the AAV-hTau-treated mice in terms of several parameters, including the escape latency and percentage of time spent in the target quadrant. In addition, neither the AAV-hTau injection nor the SSc treatment affected the swimming speed of the mice in the probe trial, verifying that all the mice had no swimming issues.

Pharmacological interventions targeting tau hyperphosphorylation are believed to be effective therapeutic strategies for treating AD. Transgenic mice expressing WT hTau as well as those expressing hTau proteins with P301L or P301S mutations develop AD-like tau pathologies, including phosphorylated tau deposition and associated neurodegeneration in the brain (Jaworski et al., 2009; Takeuchi et al., 2011; Wenger et al., 2023; Wheeler et al., 2015). Induced phosphorylation of tau at Ser202, Thr205, Thr231, and Ser396 was significantly reduced by SSc treatment in the brains of the AAV-hTau mice. Since the phosphorylation of tau at Ser202, Thr205, and Thr231 is commonly related to the early stage of AD and the phosphorylation of tau at Ser396 is a hallmark of the late stage of the disease (Augustinack et al., 2002; Neddens et al., 2018), SSc may have therapeutic effects in both the initial and advanced stages of AD.

Hippocampal phosphorylated tau induces dendritic spine loss in tauopathy and AD mouse models (Kandimalla et al., 2018). To address alterations in synaptic properties in our tauopathy model, we assessed changes in dendritic spine morphology. First, the level of DCX was assessed due to its robust presence in dendrites and its role in regulating dendritic spine development (Cohen et al., 2008; Rao and Shetty, 2004). DCX immunoreactivity and expression in the hippocampal DG region were profoundly impaired in the AAV-hTau mouse brain, and these changes were alleviated by SSc treatment. Given that DCX is a microtubule marker of neurogenesis (Francis et al., 1999), SSc might relieve hippocampal neurogenesis deficits in a tauopathy mouse model. Encouraged by the effects of SSc on DCX expression, we analyzed DCX-associated dendrite maturation in the hippocampal DG region. A significant decrease in dendritic spine density was observed in the AAV-hTau-injected mice compared with the AAV-Vec-injected mice, and SSc significantly reversed this reduction to levels similar to those in the AAV-Vec-injected control group. Together with our MWM test results, these results demonstrated that SSc uptake significantly reduced tau phosphorylation, thereby restoring the dendritic spine number, which could explain the potential therapeutic effects of SSc on the behavioral deficits observed in AAV-hTau mice.

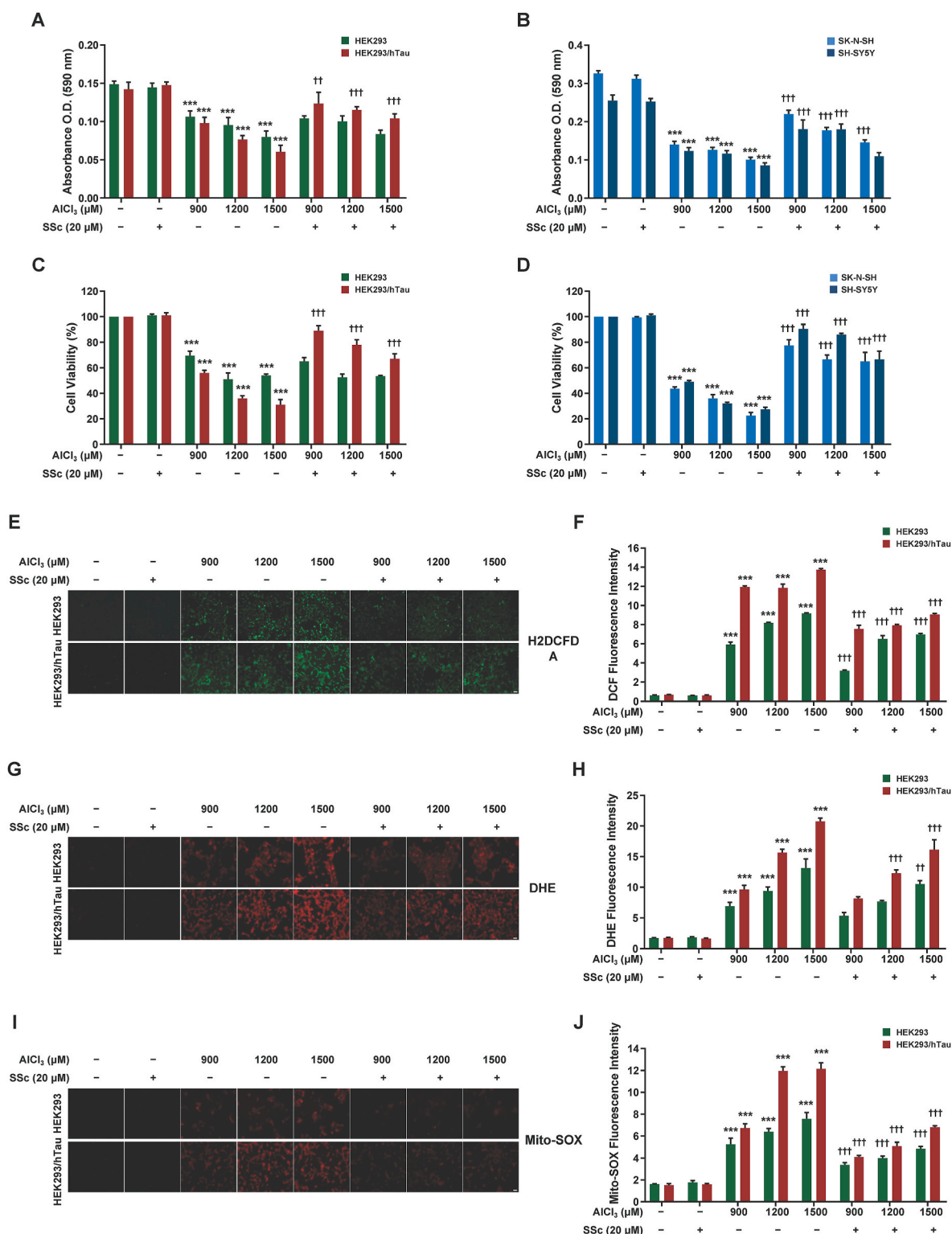


Fig. 9. SSc attenuates AlCl₃-triggered ROS generation, thereby exerting a cytoprotective effect. (A–D) HEK293, HEK293/hTau, SK-N-SH and SH-SY5Y cells were treated with 20 μM SSc for 2 h prior to cotreatment with the indicated concentrations of AlCl₃ for 48 h (for the MTT assay) (A and B) or for 72 h (for the trypan blue cell viability assay) (C and D). ****P* < 0.001 compared with untreated controls, one-way ANOVA/Dunnett's test; ††*P* < 0.01 and †††*P* < 0.001 compared with AlCl₃ alone, one-way ANOVA/Sidak's test. (E–J) HEK293 and HEK293/hTau cells were treated with 20 μM SSc for 2 h prior to cotreatment with the indicated concentrations of AlCl₃ for 6 h. Cytosolic H₂O₂ levels were detected via H2DCFDA (E–F). Cytosolic and mitochondrial superoxide levels were detected via DHE and MitoSOX, respectively (G–J). The scale bar represents 50 μm ****P* < 0.001 compared with the untreated controls, one-way ANOVA/Dunnett's test; ††*P* < 0.01 and †††*P* < 0.001 compared with AlCl₃ alone, one-way ANOVA/Sidak's test.

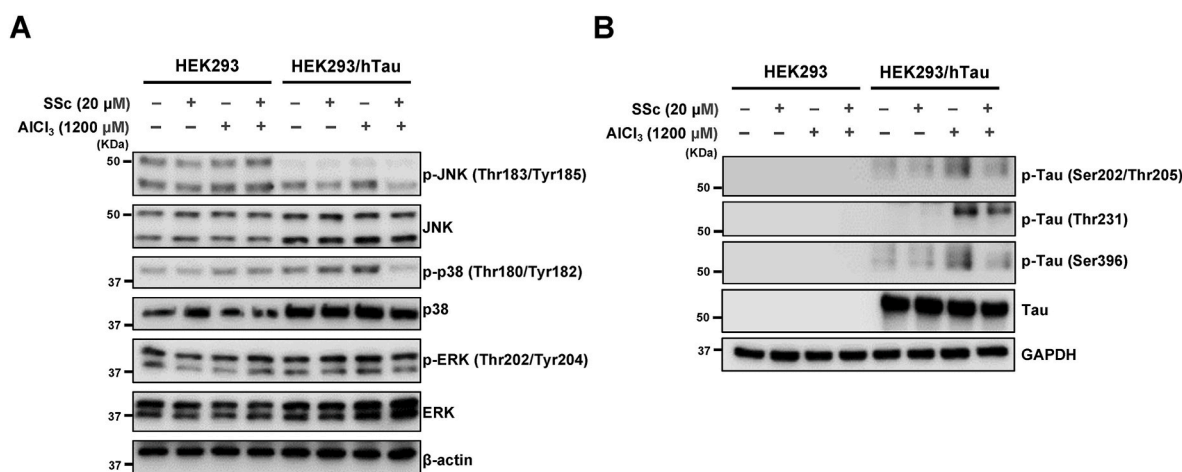


Fig. 10. SSc suppresses the AlCl₃-triggered phosphorylation of MAPKs and tau. (A) HEK293 and HEK293/hTau cells were treated with 20 μM SSc for 2 h prior to cotreatment with 1200 μM AlCl₃ for 24 h. Phospho-specific MAPK antibodies and total MAPK antibodies were used to detect phosphorylated and total MAPK protein levels, respectively. An anti-β-actin antibody was used as a loading control. (B) HEK293 and HEK293/hTau cells were treated with 20 μM SSc for 2 h prior to cotreatment with 1200 μM AlCl₃ for 48 h. Phospho-specific tau antibodies and Tau-5 were used to detect phosphorylated and total tau protein levels. An anti-GAPDH antibody was used as a loading control.

GFAP and Iba1, which are hallmarks of reactive astrocytes and microglia, respectively, are well accepted as markers of neuroinflammation (Ahmed et al., 2007; Norden et al., 2016; Takizawa et al., 2008). The immunoreactivity of GFAP-positive astrocytes and Iba1-positive microglia was increased in the brains of AAV-hTau-treated mice, and the immunoreactivity of both cell types was significantly decreased after SSc treatment. These findings suggest that SSc can regulate both nonneuronal cells and neuronal cells. The possible cross-talk between the SSc-mediated inhibition of neuroinflammation and the inhibition of tau hyperphosphorylation warrants further investigation.

Based on behavioral and histopathological outcomes, we performed RNA sequencing to compare gene expression profiles among the Vec-con, hTau-con, and hTau_SSc groups, aiming to elucidate molecular signaling mechanisms modulated by hTau injection and SSc treatment. Hierarchical clustering and heatmap analysis of DEGs revealed distinct gene expression patterns in response to hTau pathology and SSc intervention. Subsequent GSEA identified key biological processes and pathways, including oxidative phosphorylation-mediated oxidative stress and dendritic spine development, as potential therapeutic targets of SSc. The enrichment map further highlighted several critical signaling pathways associated with AAV-hTau-induced tauopathy and SSc treatment, notably the MAPK, Rho, oxidative phosphorylation, and neurotransmitter regulation pathways. Considering that oxidative phosphorylation contributes to both oxidative stress generation and antioxidant defense mechanisms (Khan et al., 2016), the observed upregulation of oxidative phosphorylation genes in the hTau_SSc group compared to the hTau-con group may indicate activation of SSc-mediated antioxidant responses. Moreover, SSc has been reported to directly bind to human serum albumin and NADPH oxidase 5, both of which are associated with oxidative processes (Chen et al., 2016; Chikhale et al., 2021). Given the central role of the MAPK cascade in regulating tau phosphorylation and amyloid beta (Aβ) metabolism-key pathological features of AD, it is plausible that SSc exerts its neuroprotective effects through modulation of this pathway. To our knowledge, there are currently no published studies demonstrating direct molecular targets of SSc, such as direct binding to Aβ, tau, or MAPK proteins. Additionally, our study did not perform direct binding assays, including molecular docking or surface plasmon resonance, to assess potential physical interactions between SSc and these proteins. Therefore, the current evidence supports an indirect mechanism whereby SSc influences intracellular signaling networks that modulate tau phosphorylation and oxidative stress rather than direct molecular binding.

Future investigations employing protein-ligand binding studies or *in silico* modeling are warranted to further clarify the direct molecular targets of SSc, as suggested in prior literature (Chen et al., 2016; Chikhale et al., 2021). Building on these findings, we have further explored oxidative stress and MAPK pathways in our tauopathy mouse and cell culture models to delineate the mechanistic basis of therapeutic effects by SSc.

Oxidative stress, which usually occurs early in the progression of AD, plays an important role in the pathogenesis of AD. An impaired antioxidant system can disrupt redox balance, triggering an oxidative state during the progression of AD-like damage. In particular, increased oxidative stress is correlated with tau pathology (Alavi Naini and Soussi-Yanicostas, 2015). Given the decrease in DT levels and increase in SOD2 levels caused by SSc treatment in AAV-hTau mice, antioxidant actions might underlie the therapeutic effects of SSc on AD. We used aluminum to investigate the protective effects of SSc on oxidative stress and tau pathology because aluminum intake can induce oxidative stress and tau hyperphosphorylation, thereby increasing the risk of developing AD (Dey and Singh, 2022; Skalny et al., 2021; Walton, 2007; Yu et al., 2014; Zhao et al., 2020). We used HEK293 and hTau-overexpressing HEK293 (HEK293/hTau) cells to evaluate the involvement of tau in AlCl₃-induced oxidative stress and cytotoxicity and the protective effects of SSc. We also used two neuroblastoma cell lines, SH-SY5Y and SK-N-SH, with endogenous expression of tau. Interestingly, the SSc-mediated protective effects on AlCl₃-induced oxidative stress and cytotoxicity were highly dependent not only on exogenous but also on endogenous tau expression. Since oxidative stress is essential for aluminum-induced cytotoxicity (Dey and Singh, 2022; Skalny et al., 2021; Walton, 2007; Yu et al., 2014; Zhao et al., 2020) and can trigger tau modifications in tauopathy (Alavi Naini and Soussi-Yanicostas, 2015), we investigated the role of oxidative stress in aluminum-triggered tau toxicity. Our results obtained via the use of two ROS scavengers, NAC and tempol, suggested that aluminum-induced ROS generation is a prerequisite for the aluminum-induced abnormal phosphorylation of tau and is a potential therapeutic target for SSc in the treatment of tau pathology. Moreover, in addition to NAC, tempol, a superoxide dismutase mimetic agent, attenuated AlCl₃-induced ROS generation and subsequent tau phosphorylation. Since the dismutase activity of tempol can catalyze the reaction between superoxide anions (O₂^{•-}) and hydrogen (H⁺) to form hydrogen peroxide (H₂O₂) and oxygen (2O₂^{•-} + 2H⁺ → H₂O₂ + O₂), superoxide anions, as well as H₂O₂, might be responsible for AlCl₃-induced oxidative damage and tau

phosphorylation and might be target molecules of SSc-mediated protection. Furthermore, AlCl_3 -induced mitochondrial superoxide production, as visualized by MitoSOX, and AlCl_3 -induced cytosolic H_2O_2 production, as visualized by DCFDA, were decreased by SSc treatment, indicating that SSc is a mitochondrion-targeted ROS scavenger. Overall, SSc seems to have protective effects on AlCl_3 -induced injury by reducing ROS generation.

Dysregulation of the MAPK signaling pathway has been implicated in AD. Although the involvement of ERK in tauopathy is debated, both JNK and p38 have been shown to be deregulated in a mouse model of tauopathy (Bendotti et al., 2006; Munoz and Ammit, 2010; Savage et al., 2002; Yarza et al., 2016), indicating that they are valid therapeutic targets for the treatment of tauopathies. Our experiments revealed increased phosphorylation of JNK, p38, and ERK in AAV-hTau mice, and all of these increases were significantly decreased following SSc treatment. The same effects of SSc were observed in HEK293/hTau cells after AlCl_3 treatment. Moreover, time course analysis after treatment with AlCl_3 , a tauopathy inducer, and treatment with MAPK inhibitors revealed that MAPK pathway activation is a prerequisite for AlCl_3 -induced abnormal tau phosphorylation. We found that AlCl_3 -induced MAPK activation is negatively regulated by ROS scavenging. These results suggest that SSc-mediated antioxidant properties and inhibition of MAPK signaling are upstream events by which SSc exerts protective effects on tau pathologies. Additionally, we explored the protective effects of SSc along with ROS scavengers and MAPK inhibitors on the AlCl_3 -induced abnormal phosphorylation of tau at multiple AD-related sites (Supplementary Fig. S11).

Despite the significant protective effects of SSc on abnormal tau phosphorylation, the effects of SSc on advanced tauopathy phenotypes, such as neurofibrillary tau tangles and neuronal loss, remain uncertain. In fact, neurofibrillary tau tangles and neuronal loss could not be assayed in our AAV-hTau mouse model. Because mutated tau shows more aggressive tau modification and forms more abundant tau filaments than does WT tau, further investigation into the inhibitory effects of SSc on tau pathology using an AAV vector encoding a hTau protein with a P301 mutation, such as P301L or P301S, is necessary. The significant effects of SSc on tauopathy substantiate the need for investigation into the inhibitory effects of SSc on amyloidosis because SSc was shown to target $\text{A}\beta$ as well as tau in neuronal cell models in our previous study (Lee et al., 2016). In fact, studies regarding the possible therapeutic activities of SSc against amyloidosis are in progress.

This study has several limitations. First, we exclusively used male mice in our AD model to reduce biological variability associated with sex hormones, particularly the estrous cycle in females, which is known to influence performance in cognitive behavioral tests. However, this approach may overlook important sex-specific differences in disease pathology and treatment response, thereby introducing translational bias and limiting the generalizability of the findings to human populations. Given the well-established sex differences in AD, future studies should include female mice to enable a comprehensive evaluation of sex-dependent effects. Second, the study relied solely on the MWM to assess cognitive function, which restricts the evaluation to spatial learning and memory. This limited behavioral assessment may not capture broader cognitive domains, thereby reducing the translational relevance to human cognitive impairment. Incorporating additional behavioral paradigms, such as the Y-maze, absent field, and fear conditioning tests, would provide a more comprehensive characterization of the neurocognitive effects of SSc. Finally, the study did not quantitatively assess the BBB permeability of SSc under dietary administration conditions. Future investigations should include quantitative evaluations of BBB penetration following oral intake via a normal diet to determine the optimal dosage and bioavailability of SSc in the central nervous system.

5. Conclusion

Our study was conducted to investigate the therapeutic potential of

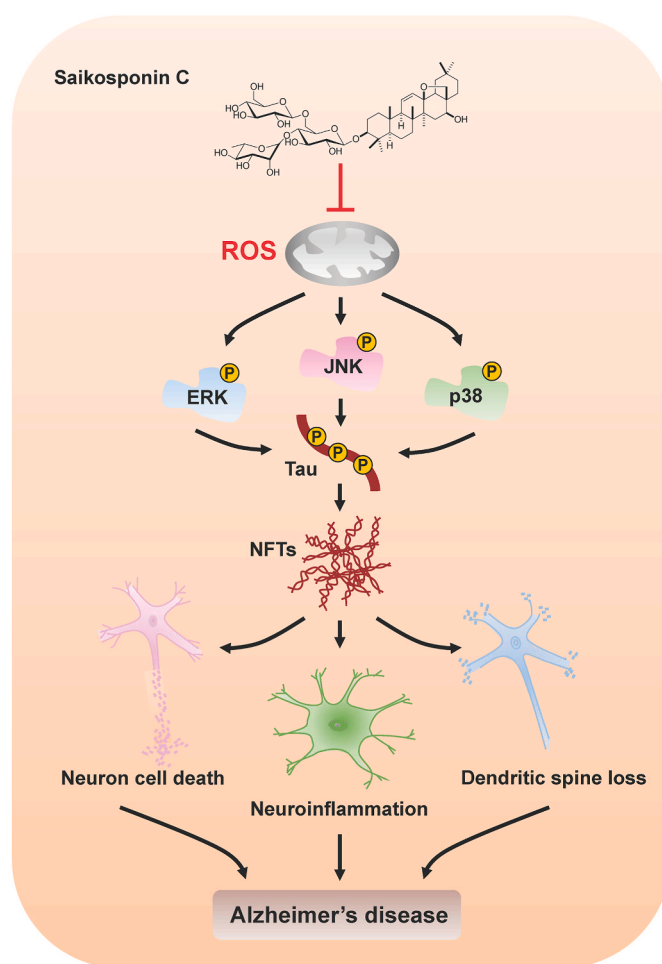


Fig. 11. SSc attenuates tauopathy via the ROS/MAPK axis in AD. SSc decreases the level of ROS, which inhibits the activity of MAPKs, including ERK, JNK and p38. Furthermore, the inactivation of MAPKs ultimately alleviates tau hyperphosphorylation and tauopathy in the AD brain.

SSc against AD in terms of tau pathology in an animal model. Our research revealed that SSc exerted anti-AD effects in an *in vivo* AAV-based tauopathy model. We propose that the oral administration of SSc could be a promising therapeutic strategy for treating tauopathies in clinical trials. Mechanistically, our findings indicate that SSc can exert anti-tau effects through the inhibition of ROS generation, the inhibition of the MAPK pathway and the subsequent suppression of tau phosphorylation in animal and cell culture models (Fig. 11).

CRediT authorship contribution statement

Dongmei Chen: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Ruomeng Li:** Methodology, Investigation, Formal analysis, Data curation. **Fei She:** Investigation, Formal analysis, Data curation. **Jiawen Chen:** Investigation, Data curation. **Sang Min Cho:** Investigation, Formal analysis. **Bohyun Yang:** Investigation, Formal analysis, Data curation. **Yuan Tian:** Resources, Methodology. **Le Guan:** Resources, Methodology. **Xiuzhi Zheng:** Validation, Methodology. **Quling Wang:** Resources, Methodology. **Long Wang:** Resources, Methodology. **Sang Won Park:** Writing – review & editing, Resources, Methodology. **Wucheng Tao:** Writing – review & editing, Resources, Methodology. **Tao Zhang:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Byeong Mo Kim:** Writing – review & editing, Writing – original draft,

Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tae Ho Lee:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethnic statement

All animal experiments were approved by the Experimental Animal Ethics Committee of Fujian Medical University (IACUC FJMU, 2023-Y-0360, March 3, 2023).

Data availability

The RNA-seq data reported in this article have been deposited in the GSA and are accessible through the GSA series accession number [CRA025339](#). Other data will be made available on request.

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Declaration of competing interest

Byeong Mo Kim is a founder of and owns equity in AgingTarget Inc. Other authors do not have any competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.120221>.

Abbreviations

AAV	Adeno-associated virus
AD	Alzheimer's disease
AlCl ₃	Aluminum chloride
CA3	Cornu ammonis 3
DCX	Doublecortin
EMEM	Eagle's minimum essential medium
FBS	Fetal bovine serum
FPKM	Fragments per kilobase of exon per million reads
GSEA	Gene set enrichment analysis
HRP	Horseradish peroxidase
hTau	Human tau
MAPK	Mitogen-activated protein kinase
MCI	Mild cognitive impairment
MWM	Morris water maze
NES	Normalized enrichment score
ROS	Reactive oxygen species
RQN	RNA quality number
SOD2	Superoxide dismutase 2
SSc	Saikosaponin C

SEMs Standard errors of the means
Tauopathy Tau-related pathology
WT wild-type

Data availability

Data will be made available on request.

References

- Ahmed, Z., Shaw, G., Sharma, V.P., Yang, C., McGowan, E., Dickson, D.W., 2007. Actin-binding proteins coronin-1a and IBA-1 are effective microglial markers for immunohistochemistry. *J. Histochem. Cytochem.* 55 (7), 687–700.
- Alavi Naini, S.M., Soussi-Yanicostas, N., 2015. Tau hyperphosphorylation and oxidative stress, a critical vicious circle in neurodegenerative tauopathies? *Oxid. Med. Cell. Longev.* 2015, 151979.
- Andorfer, C., Kress, Y., Espinoza, M., de Silva, R., Tucker, K.L., Barde, Y.A., Duff, K., Davies, P., 2003. Hyperphosphorylation and aggregation of tau in mice expressing normal human tau isoforms. *J. Neurochem.* 86 (3), 582–590.
- Augustinack, J.C., Schneider, A., Mandelkow, E.M., Hyman, B.T., 2002. Specific tau phosphorylation sites correlate with severity of neuronal cytopathology in Alzheimer's disease. *Acta Neuropathol.* 103 (1), 26–35.
- Bai, R., Guo, J., Ye, X.Y., Xie, Y., Xie, T., 2022. Oxidative stress: the core pathogenesis and mechanism of Alzheimer's disease. *Ageing Res. Rev.* 77, 101619.
- Bai, Z., Gao, T., Zhang, R., Lu, Y., Tian, J., Wang, T., Zhao, K., Wang, H., 2023. Inhibition of IL-6 methylation by Saikosaponin C regulates neuroinflammation to alleviate depression. *Int. Immunopharmacol.* 118, 110043.
- Bejanin, A., Schonhaut, D.R., La Joie, R., Kramer, J.H., Baker, S.L., Sosa, N., Ayakta, N., Cantwell, A., Janabi, M., Lauriola, M., O'Neil, J.P., Gorno-Tempini, M.L., Miller, Z. A., Rosen, H.J., Miller, B.L., Jagust, W.J., Rabinovici, G.D., 2017. Tau pathology and neurodegeneration contribute to cognitive impairment in Alzheimer's disease. *Brain* 140 (12), 3286–3300.
- Bendotti, C., Tortarolo, M., Borsello, T., 2006. Targeting stress activated protein kinases, JNK and p38, as new therapeutic approach for neurodegenerative diseases. *Cent. Nerv. Syst. Agents Med. Chem.* 6 (2), 109–117.
- Bramblett, G.T., Goedert, M., Jakes, R., Merrick, S.E., Trojanowski, J.Q., Lee, V.M., 1993. Abnormal tau phosphorylation at Ser396 in Alzheimer's disease recapitulates development and contributes to reduced microtubule binding. *Neuron* 10 (6), 1089–1099.
- Buerger, K., Zinkowski, R., Teipel, S.J., Tapiola, T., Arai, H., Blennow, K., Andreasen, N., Hofmann-Kiefer, K., DeBernardis, J., Kerkman, D., McCulloch, C., Kohnen, R., Padberg, F., Pirttilä, T., Schapiro, M.B., Rapoport, S.I., Moller, H.J., Davies, P., Hampel, H., 2002. Differential diagnosis of alzheimer disease with cerebrospinal fluid levels of tau protein phosphorylated at threonine 231. *Arch. Neurol.* 59 (8), 1267–1272.
- Campos, H.M., da Costa, M., da Silva Moreira, L.K., da Silva Neri, H.F., Branco da Silva, C.R., Pruccoli, L., Dos Santos, F.C.A., Costa, E.A., Tarozzi, A., Ghedini, P.C., 2022. Protective effects of chrysin against the neurotoxicity induced by aluminium: in vitro and in vivo studies. *Toxicology* 465, 153033.
- Chen, D., Lan, G., Li, R., Mei, Y., Shui, X., Gu, X., Wang, L., Zhang, T., Gan, C.L., Xia, Y., Hu, L., Tian, Y., Zhang, M., Lee, T.H., 2022. Melatonin ameliorates tau-related pathology via the miR-504-3p and CDK5 axis in Alzheimer's disease. *Transl. Neurodegener.* 11 (1), 27.
- Chen, D., Mei, Y., Kim, N., Lan, G., Gan, C.L., Fan, F., Zhang, T., Xia, Y., Wang, L., Lin, C., Ke, F., Zhou, X.Z., Lu, K.P., Lee, T.H., 2020. Melatonin directly binds and inhibits death-associated protein kinase 1 function in Alzheimer's disease. *J. Pineal Res.* 69 (2), e12665.
- Chen, Y.C., Wang, H.M., Niu, Q.X., Ye, D.Y., Liang, G.W., 2016. Binding between saikosaponin C and human serum albumin by fluorescence spectroscopy and molecular docking. *Molecules* 21 (2), 153.
- Chikhale, R., Sinha, S.K., Wanjari, M., Gurav, N.S., Ayyanar, M., Prasad, S., Khanal, P., Dey, Y.N., Patil, R.B., Gurav, S.S., 2021. Computational assessment of saikosaponins as adjuvant treatment for COVID-19: molecular docking, dynamics, and network pharmacology analysis. *Mol. Divers.* 25 (3), 1889–1904.
- Cohen, D., Segal, M., Reiner, O., 2008. Doublecortin supports the development of dendritic arbors in primary hippocampal neurons. *Dev. Neurosci.* 30 (1–3), 187–199.
- Denechaud, M., Geurs, S., Comptdaer, T., Begard, S., Garcia-Nunez, A., Pechereau, L.A., Bouillet, T., Vermeiren, Y., De Deyn, P.P., Perbet, R., Deramecourt, V., Maurage, C. A., Vanderhaegen, M., Vanuytven, S., Lefebvre, B., Bogaert, E., Deglon, N., Voet, T., Colin, M., Buee, L., Dermaut, B., Galas, M.C., 2023. Tau promotes oxidative stress-associated cycling neurons in S phase as a pro-survival mechanism: possible implication for Alzheimer's disease. *Prog. Neurobiol.* 223, 102386.
- Dey, M., Singh, R.K., 2022. Chronic oral exposure of aluminum chloride in rat modulates molecular and functional neurotoxic markers relevant to Alzheimer's disease. *Toxicol. Mech. Methods* 32 (8), 616–627.
- Duff, K., Knight, H., Refolo, L.M., Sanders, S., Yu, X., Picciano, M., Malester, B., Hutton, M., Adamson, J., Goedert, M., Burki, K., Davies, P., 2000. Characterization of pathology in transgenic mice over-expressing human genomic and cDNA tau transgenes. *Neurobiol. Dis.* 7 (2), 87–98.
- Feng, Y., Gao, X., Meng, M., Xue, H., Qin, X., 2020. Multi-omics reveals the mechanisms of antidepressant-like effects of the low polarity fraction of bupleuri radix. *J. Ethnopharmacol.* 256, 112806.

- Ferrer, I., Gomez-Isla, T., Puig, B., Freixes, M., Ribe, E., Dalfó, E., Avila, J., 2005. Current advances on different kinases involved in tau phosphorylation, and implications in Alzheimer's disease and tauopathies. *Curr. Alzheimer Res.* 2 (1), 3–18.
- Fisar, Z., Hroudova, J., 2024. CoQ(10) and mitochondrial dysfunction in Alzheimer's disease. *Antioxidants* 13 (2).
- Francis, F., Koulakoff, A., Boucher, D., Chafey, P., Schaar, B., Vinet, M.C., Friocourt, G., McDonnell, N., Reiner, O., Kahn, A., McConnell, S.K., Berwald-Netter, Y., Denoulet, P., Chelly, J., 1999. Doublecortin is a developmentally regulated, microtubule-associated protein expressed in migrating and differentiating neurons. *Neuron* 23 (2), 247–256.
- Fu, L., Shi, S., Yi, J., Wang, N., He, Y., Wu, Z., Peng, J., Deng, Y., Wang, W., Wu, C., Lyu, A., Zeng, X., Zhao, W., Hou, T., Cao, D., 2024. ADMETlab 3.0: an updated comprehensive online ADMET prediction platform enhanced with broader coverage, improved performance, API functionality and decision support. *Nucleic Acids Res.* 52 (W1), W422–W431.
- Gan, C.L., Zou, Y., Chen, D., Shui, X., Hu, L., Li, R., Zhang, T., Wang, J., Mei, Y., Wang, L., Zhang, M., Tian, Y., Gu, X., Lee, T.H., 2022. Blocking ERK-DAPK1 axis attenuates glutamate excitotoxicity in epilepsy. *Int. J. Mol. Sci.* 23 (12).
- Gao, X., Li, Y., Meng, M., Wang, P., Feng, Y., Jia, J., Qin, X., 2020. Exploration of chemical composition and absorption characteristics of Chaigui granules based on UHPLC-Q-orbitrap-MS/MS. *J. Pharm. Biomed. Anal.* 187, 113293.
- Goedert, M., Spillantini, M.G., Cairns, N.J., Crowther, R.A., 1992. Tau proteins of alzheimer paired helical filaments: abnormal phosphorylation of all six brain isoforms. *Neuron* 8 (1), 159–168.
- Han, X., Ma, D., Wang, J., Pei, L., Liu, L., Shi, W., Rong, Z., Wang, X., Zhang, Y., Zheng, Y., Sun, H., 2024. Spatial mapping of bioactive metabolites in the roots of three bupleurum species by matrix-assisted laser desorption/ionization mass spectrometry imaging. *Molecules* 29 (16).
- Heneka, M.T., Carson, M.J., El Khoury, J., Landreth, G.E., Brosseron, F., Feinstein, D.L., Jacobs, A.H., Wyss-Coray, T., Vitorica, J., Ransohoff, R.M., Herrup, K., Frautschy, S. A., Finsen, B., Brown, G.C., Verkhratsky, A., Yamanaka, K., Koistinaho, J., Latz, E., Halle, A., Petzold, G.C., Town, T., Morgan, D., Shinohara, M.L., Perry, V.H., Holmes, C., Bazan, N.G., Brooks, D.J., Hunot, S., Joseph, B., Deigendesch, N., Garaschuk, O., Boddeke, E., Dinarello, C.A., Breitner, J.C., Cole, G.M., Golenbock, D. T., Kummer, M.P., 2015. Neuroinflammation in Alzheimer's disease. *Lancet Neurol.* 14 (4), 388–405.
- Huang, L.K., Kuan, Y.C., Lin, H.W., Hu, C.J., 2023. Clinical trials of new drugs for alzheimer disease: a 2020–2023 update. *J. Biomed. Sci.* 30 (1), 83.
- Ionescu-Tucker, A., Cotman, C.W., 2021. Emerging roles of oxidative stress in brain aging and Alzheimer's disease. *Neurobiol. Aging* 107, 86–95.
- Iqbal, K., Alonso Adel, C., Chen, S., Chohan, M.O., El-Akkad, E., Gong, C.X., Khatoon, S., Li, B., Liu, F., Rahman, A., Tanimukai, H., Grundke-Iqbal, I., 2005. Tau pathology in alzheimer disease and other tauopathies. *Biochim. Biophys. Acta* 1739 (2–3), 198–210.
- Jaworski, T., Dewachter, I., Lechat, B., Croes, S., Termont, A., Demedts, D., Borghgraef, P., Devijver, H., Filipkowski, R.K., Kaczmarek, L., Kugler, S., Van Leuven, F., 2009. AAV-tau mediates pyramidal neurodegeneration by cell-cycle re-entry without neurofibrillary tangle formation in wild-type mice. *PLoS One* 4 (10), e7280.
- Jia, A., Yang, X., Zou, B., Li, J., Wang, Y., Ma, R., Li, J., Yao, Y., 2022. Saikosaponins: a review of structures and pharmacological activities. *Nat. Prod. Commun.* 17 (5), 1–24.
- Jia, R., Gu, Z., He, Q., Du, J., Cao, L., Jeney, G., Xu, P., Yin, G., 2019. Anti-oxidative, anti-inflammatory and hepatoprotective effects of Radix bupleuri extract against oxidative damage in tilapia (*Oreochromis niloticus*) via Nrf2 and TLRs signaling pathway. *Fish Shellfish Immunol.* 93, 395–405.
- Jiang, S., Li, C., Liu, D., Zeng, F., Wei, W., He, T., Yang, W., 2025. Role, mechanisms and effects of Radix bupleuri in anti-breast cancer. *Oncol. Lett.* 29 (4), 166 (Review).
- Kandimalla, R., Manczak, M., Yin, X., Wang, R., Reddy, P.H., 2018. Hippocampal phosphorylated tau induced cognitive decline, dendritic spine loss and mitochondrial abnormalities in a mouse model of Alzheimer's disease. *Hum. Mol. Genet.* 27 (1), 30–40.
- Khan, A.U.H., Rathore, M.G., Allende-Vega, N., Vo, D.N., Belkhala, S., Orecchioni, S., Talarico, G., Bertolini, F., Cartron, G., Lecellier, C.H., Villalba, M., 2016. Human leukemic cells performing oxidative phosphorylation (OXPHOS) generate an antioxidant response independently of Reactive Oxygen species (ROS) production. *EBioMedicine* 3, 43–53.
- Lee, T.H., Chang, J., Kim, B.M., 2014. Saikosaponin C inhibits lipopolysaccharide-induced apoptosis by suppressing caspase-3 activation and subsequent degradation of focal adhesion kinase in human umbilical vein endothelial cells. *Biochem. Biophys. Res. Commun.* 445 (3), 615–621.
- Lee, T.H., Park, S., You, M.H., Lim, J.H., Min, S.H., Kim, B.M., 2016. A potential therapeutic effect of saikosaponin C as a novel dual-target anti-alzheimer agent. *J. Neurochem.* 136 (6), 1232–1245.
- Lei, X., Lv, L., Ma, Y., Zhang, C., Li, L., Huang, Y., 2025. Investigation of potential targets for bupleuri radix in the treatment of subacute thyroiditis. *Internet J. Endocrinol.* 2025, 7460422.
- Li, J., Zou, B., Cheng, X.Y., Yang, X.H., Li, J., Zhao, C.H., Ma, R.X., Tian, J.X., Yao, Y., 2022. Therapeutic effects of total saikosaponins from Radix bupleuri against Alzheimer's disease. *Front. Pharmacol.* 13, 940999.
- Li, R., Zhi, S., Lan, G., Chen, X., Zheng, X., Hu, L., Wang, L., Zhang, T., Lee, T.H., Rao, S., Chen, D., 2023. Ablation of death-associated protein kinase 1 changes the transcriptomic profile and alters neural-related pathways in the brain. *Int. J. Mol. Sci.* 24 (7).
- Liberzon, A., Subramanian, A., Pinchback, R., Thorvaldsdottir, H., Tamayo, P., Mesirov, J.P., 2011. Molecular signatures database (MSigDB) 3.0. *Bioinformatics* 27 (12), 1739–1740.
- Liu, X.A., Song, J., Jiang, Q., Wang, Q., Tian, Q., Wang, J.Z., 2012. Expression of the hyperphosphorylated tau attenuates ER stress-induced apoptosis with upregulation of unfolded protein response. *Apoptosis* 17 (10), 1039–1049.
- Mecocci, P., Boccardi, V., Cecchetti, R., Bastiani, P., Scamosci, M., Ruggiero, C., Baroni, M., 2018. A long journey into aging, brain aging, and Alzheimer's disease following the oxidative stress tracks. *J. Alzheimers Dis.* 62 (3), 1319–1335.
- Mei, Y., She, F., Zhang, L., Kim, G., Li, R., Zheng, X., Wang, Z., Chen, R., Wang, L., Chen, D., Kim, J., Zhang, T., Lee, T.H., 2025. Zipper-interacting protein kinase mediates neuronal cell death and cognitive dysfunction in traumatic brain injury via regulating DEDD. *Cell Death Dis.* 16 (1), 151.
- Merico, D., Isserlin, R., Stueker, O., Emili, A., Bader, G.D., 2010. Enrichment map: a network-based method for gene-set enrichment visualization and interpretation. *PLoS One* 5 (11), e13984.
- Morris, M., Maeda, S., Vossel, K., Mucke, L., 2011. The many faces of tau. *Neuron* 70 (3), 410–426.
- Mu, Y., Gage, F.H., 2011. Adult hippocampal neurogenesis and its role in Alzheimer's disease. *Mol. Neurodegener.* 6, 85.
- Munoz, L., Ammit, A.J., 2010. Targeting p38 MAPK pathway for the treatment of Alzheimer's disease. *Neuropharmacology* 58 (3), 561–568.
- Neddens, J., Temmel, M., Flunkert, S., Kerschbaumer, B., Hoeller, C., Loeffler, T., Niederkofler, V., Daum, G., Attems, J., Hutter-Paier, B., 2018. Phosphorylation of different tau sites during progression of Alzheimer's disease. *Acta Neuropathol. Commun.* 6 (1), 52.
- Ni, L.J., Zhang, L.G., Hou, J., Shi, W.Z., Guo, M.L., 2009. A strategy for evaluating antipyretic efficacy of Chinese herbal medicines based on UV spectra fingerprints. *J. Ethnopharmacol.* 124 (1), 79–86.
- Norden, D.M., Trojanowski, P.J., Villanueva, E., Navarro, E., Godbout, J.P., 2016. Sequential activation of microglia and astrocyte cytokine expression precedes increased Iba-1 or GFAP immunoreactivity following systemic immune challenge. *Glia* 64 (2), 300–316.
- Okita, K., Li, Q., Murakami, T., Takahashi, M., 1993. Anti-growth effects with components of sho-saiko-to (TJ-9) on cultured human hepatoma cells. *Eur. J. Cancer Prev.* 2 (2), 169–175.
- Pan, Y., Ke, Z., Ye, H., Sun, L., Ding, X., Shen, Y., Zhang, R., Yuan, J., 2019. Saikosaponin C exerts anti-HBV effects by attenuating HNF1alpha and HNF4alpha expression to suppress HBV pgRNA synthesis. *Inflamm. Res.* 68 (12), 1025–1034.
- Promyo, K., Iqbal, F., Chaidze, N., Chetsawang, B., 2020. Aluminum chloride-induced amyloid beta accumulation and endoplasmic reticulum stress in rat brain are averted by melatonin. *Food Chem. Toxicol.* 146, 111829.
- Qu, X., Hu, S., Li, T., Zhang, J., Wang, B., Liu, C., 2022. Metabolomics analysis reveals the differences between Bupleurum chinense DC. and Bupleurum scorzonifolium willd. *Front. Plant Sci.* 13, 933849.
- Rao, M.S., Shetty, A.K., 2004. Efficacy of doublecortin as a marker to analyse the absolute number and dendritic growth of newly generated neurons in the adult dentate gyrus. *Eur. J. Neurosci.* 19 (2), 234–246.
- Rawat, P., Sehara, U., Bisht, J., Selman, A., Culbertson, J., Reddy, P.H., 2022. Phosphorylated Tau in Alzheimer's Disease and other tauopathies. *Int. J. Mol. Sci.* 23 (21).
- Rui, T.Y., Huang, H.Z., Zheng, K., Fan, H.W., Zhang, J., Guo, Z.Y., Man, H.Y., Brazhe, N., Semyanov, A., Lu, Y.M., Liu, D., Zhu, L.Q., 2025. Tau pathology drives disease-associated astrocyte reactivity in salt-induced neurodegeneration. *Adv. Sci. (Weinh.)* 12 (11), e2410799.
- Savage, M.J., Lin, Y.G., Ciallella, J.R., Flood, D.G., Scott, R.W., 2002. Activation of c-Jun N-terminal kinase and p38 in an Alzheimer's disease model is associated with amyloid deposition. *J. Neurosci.* 22 (9), 3376–3385.
- Shannon, P., Markiel, A., Ozier, O., Baliga, N.S., Wang, J.T., Ramage, D., Amin, N., Schwikowski, B., Ideker, T., 2003. Cytoscape: a software environment for integrated models of biomolecular interaction networks. *Genome Res.* 13 (11), 2498–2504.
- Shyu, K.G., Tsai, S.C., Wang, B.W., Liu, Y.C., Lee, C.C., 2004. Saikosaponin C induces endothelial cells growth, migration and capillary tube formation. *Life Sci.* 76 (7), 813–826.
- Singh, A., Ansari, V.A., Mahmood, T., Hasan, S.M., Wasim, R., Maheshwari, S., Akhtar, J., Sheikh, S., Vishwakarma, V.K., 2024. Targeting abnormal tau phosphorylation for Alzheimer's therapeutics. *Horm. Metab. Res.* 56 (7), 482–488.
- Skalny, A.V., Aschner, M., Jiang, Y., Gluhcheva, Y.G., Tizabi, Y., Lobinski, R., Tinkov, A. A., 2021. Molecular mechanisms of aluminum neurotoxicity: update on adverse effects and therapeutic strategies. *Adv. Neurotoxicol.* 5, 1–34.
- Subramanian, A., Tamayo, P., Mootha, V.K., Mukherjee, S., Ebert, B.L., Gillette, M.A., Paulovich, A., Pomeroy, S.L., Golub, T.R., Lander, E.S., Mesirov, J.P., 2005. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc. Natl. Acad. Sci. U. S. A.* 102 (43), 15545–15550.
- Takeuchi, H., Iba, M., Inoue, H., Higuchi, M., Takao, K., Tsukita, K., Karatsu, Y., Iwamoto, Y., Miyakawa, T., Suhara, T., Trojanowski, J.Q., Lee, V.M., Takahashi, R., 2011. P301S mutant human tau transgenic mice manifest early symptoms of human tauopathies with dementia and altered sensorimotor gating. *PLoS One* 6 (6), e21050.
- Takizawa, T., Gudla, P.R., Guo, L., Lockett, S., Misteli, T., 2008. Allele-specific nuclear positioning of the monoallelically expressed astrocyte marker GFAP. *Genes Dev.* 22 (4), 489–498.
- Walton, J.R., 2007. An aluminum-based rat model for Alzheimer's disease exhibits oxidative damage, inhibition of PP2A activity, hyperphosphorylated tau, and granulovacuolar degeneration. *J. Inorg. Biochem.* 101 (9), 1275–1284.
- Wang, L., Shui, X., Mei, Y., Xia, Y., Lan, G., Hu, L., Zhang, M., Gan, C.L., Li, R., Tian, Y., Wang, Q., Gu, X., Chen, D., Zhang, T., Lee, T.H., 2022a. miR-143-3p inhibits

- aberrant tau phosphorylation and amyloidogenic processing of APP by directly targeting DAPK1 in Alzheimer's disease. *Int. J. Mol. Sci.* 23 (14).
- Wang, L., Shui, X., Zhang, M., Mei, Y., Xia, Y., Lan, G., Hu, L., Gan, C.L., Tian, Y., Li, R., Gu, X., Zhang, T., Chen, D., Lee, T.H., 2022b. MiR-191-5p attenuates Tau phosphorylation, abeta generation, and neuronal cell death by regulating death-associated protein kinase 1. *ACS Chem. Neurosci.* 13 (24), 3554–3566.
- Wang, X., Li, S., Yu, J., Wang, W., Du, Z., Gao, S., Ma, Y., Tang, R., Liu, T., Ma, S., Fu, Q., Deng, X., 2023. Saikosaponin B2 ameliorates depression-induced microglia activation by inhibiting ferroptosis-mediated neuroinflammation and ER stress. *J. Ethnopharmacol.* 316, 116729.
- Wang, Y., Mandelkow, E., 2016. Tau in physiology and pathology. *Nat. Rev. Neurosci.* 17 (1), 5–21.
- Wegmann, S., Biernat, J., Mandelkow, E., 2021. A current view on Tau protein phosphorylation in Alzheimer's disease. *Curr. Opin. Neurobiol.* 69, 131–138.
- Wenger, K., Viode, A., Schlaffner, C.N., van Zalm, P., Cheng, L., Dellovade, T., Langlois, X., Bannon, A., Chang, R., Connors, T.R., Oakley, D., Renard, B., Rappsilber, J., Hyman, B., Steen, H., Steen, J.A., 2023. Common mouse models of tauopathy reflect early but not late human disease. *Mol. Neurodegener.* 18 (1), 10.
- Wheeler, J.M., McMillan, P.J., Hawk, M., Iba, M., Robinson, L., Xu, G.J., Dombroski, B. A., Jeong, D., Dichter, M.A., Juul, H., Loomis, E., Raskind, M., Leverenz, J.B., Trojanowski, J.Q., Lee, V.M., Schellenberg, G.D., Kraemer, B.C., 2015. High copy wildtype human 1N4R tau expression promotes early pathological tauopathy accompanied by cognitive deficits without progressive neurofibrillary degeneration. *Acta Neuropathol. Commun.* 3, 33.
- Xu, C., Guo, J., Li, L., Wang, X., Zhou, Q., Sun, D., Zhang, S., Li, S., Ye, J., Liu, Y., Liu, E., Zeng, P., Wang, X., Yang, Y., Wang, J.Z., 2020. Co-Expression of three wild-type 3R-Tau isoforms induces memory deficit via oxidation-related DNA damage and cell death: a promising model for tauopathies. *J. Alzheimers Dis.* 73 (3), 1105–1123.
- Yan, X., Yang, Y., Huang, W., Fu, S., Cui, B., Chu, M., Dong, Y., Peng, Y., Song, H., Shi, J., Liu, Q., 2025. Beneficial effects of the herbal medicine zuo gui wan in a mice model of Alzheimer's disease via Drp1-Mediated inhibition of mitochondrial fission and activation of AMPK/PGC-1alpha-regulated mitochondrial bioenergetics. *J. Ethnopharmacol.* 342, 119425.
- Yang, F., Dong, X., Yin, X., Wang, W., You, L., Ni, J., 2017. Radix bupleuri: a review of traditional uses, botany, phytochemistry, pharmacology, and toxicology. *Biomed. Res. Int.* 2017, 7597596.
- Yang, P., Huang, S., Luo, Z., Zhou, S., Zhang, C., Zhu, Y., Yang, J., Li, L., 2024. Radix bupleuri aqueous extract attenuates MK801-induced schizophrenia-like symptoms in mice: participation of intestinal flora. *Biomed. Pharmacother.* 172, 116267.
- Yarza, R., Vela, S., Solas, M., Ramirez, M.J., 2016. c-Jun N-terminal kinase (JNK) signaling as a therapeutic target for Alzheimer's disease. *Front. Pharmacol.* 6, 321.
- Yin, Y., Gao, D., Wang, Y., Wang, Z.H., Wang, X., Ye, J., Wu, D., Fang, L., Pi, G., Yang, Y., Wang, X.C., Lu, C., Ye, K., Wang, J.Z., 2016. Tau accumulation induces synaptic impairment and memory deficit by calcineurin-mediated inactivation of nuclear CaMKIV/CREB signaling. *Proc. Natl. Acad. Sci. U. S. A.* 113 (26), E3773–E3781.
- Yu, L., Jiang, R., Su, Q., Yu, H., Yang, J., 2014. Hippocampal neuronal metal ion imbalance related oxidative stress in a rat model of chronic aluminum exposure and neuroprotection of meloxicam. *Behav. Brain Funct.* 10, 6.
- Zhang, M., Shui, X., Zheng, X., Lee, J.E., Mei, Y., Li, R., Tian, Y., Zheng, X., Wang, Q., Wang, L., Chen, D., Zhang, T., Kim, B.M., Kim, J., Lee, T.H., 2023. Death-associated protein kinase 1 phosphorylates MDM2 and inhibits its protein stability and function. *Arch. Pharm. Res. (Seoul)* 46 (11–12), 882–896.
- Zhang, T., Kim, B.M., Lee, T.H., 2024a. Death-associated protein kinase 1 as a therapeutic target for Alzheimer's disease. *Transl. Neurodegener.* 13 (1), 4.
- Zhang, T., Tian, Y., Zheng, X., Li, R., Hu, L., Shui, X., Mei, Y., Wang, Q., Zhang, M., Zheng, X., Wang, L., Chen, D., Tao, W., Lee, T.H., 2024b. Activation of transient receptor potential vanilloid 1 ameliorates tau accumulation-induced synaptic damage and cognitive dysfunction via autophagy enhancement. *CNS Neurosci. Ther.* 30 (3), e14432.
- Zhang, T., Xia, Y., Hu, L., Chen, D., Gan, C.L., Wang, L., Mei, Y., Lan, G., Shui, X., Tian, Y., Li, R., Zhang, M., Lee, T.H., 2022. Death-associated protein kinase 1 mediates Abeta42 aggregation-induced neuronal apoptosis and tau dysregulation in Alzheimer's disease. *Int. J. Biol. Sci.* 18 (2), 693–706.
- Zhao, Y., Dang, M., Zhang, W., Lei, Y., Ramesh, T., Veeraraghavan, V.P., Hou, X., 2020. Neuroprotective effects of Syringic acid against aluminium chloride induced oxidative stress mediated neuroinflammation in rat model of Alzheimer's disease. *J. Funct. Foods* 71, 104009.
- Zhong, Y., Li, J., Zhu, X., Huang, N., Liu, R., Sun, R., 2024. A comprehensive review of bupleuri radix and its bioactive components: with a major focus on treating chronic liver diseases. *J. Ethnopharmacol.* 330, 118244.
- Zhou, L., Yang, C., Liu, Z., Chen, L., Wang, P., Zhou, Y., Yuan, M., Zhou, L.T., Wang, X., Zhu, L.Q., 2024. Neuroprotective effect of the traditional decoction tian-si-yin against Alzheimer's disease via suppression of neuroinflammation. *J. Ethnopharmacol.* 321, 117569.
- Zhou, L.T., Liu, D., Kang, H.C., Lu, L., Huang, H.Z., Ai, W.Q., Zhou, Y., Deng, M.F., Li, H., Liu, Z.Q., Zhang, W.F., Hu, Y.Z., Han, Z.T., Zhang, H.H., Jia, J.J., Sarkar, A.K., Sharaydeh, S., Wang, J., Man, H.Y., Schilling, M., Bertram, L., Lu, Y., Guo, Z., Zhu, L. Q., 2023. Tau pathology epigenetically remodels the neuron-glia cross-talk in Alzheimer's disease. *Sci. Adv.* 9 (16), eabq7105.